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Hard magnetic materials and rotor manufacturing

Vorlesung Produktion elektrischer Motoren und Maschinen
Lecture Production of electric motors and machines

The cost-effectiveness and efficiency of permanent magnet electrical machines are determined primarily by the magnets used.

The decision on which solenoids to use for permanent-magnet DC and synchronous machines is determined by the material, the manufacturing process, the operating conditions, the assembly process and the quality.

After attending this lecture unit, you should:

- **be able to describe the physical properties of hard magnetic materials,**
- **describe the rare earth materials market,**
- **describe the manufacturing processes of permanent magnets,**
- **summarize quality criteria and test procedures, and**
- **outline selection criteria and designs under different operating conditions and**
- **typical assembly processes for PM rotors.**

Graphic: Federal Office of Energy (Switzerland)

Graphic: MS-Schramberg

Graphic: Vacuumschmelze GmbH & Co. KG

Lecture unit 3 introduces the properties and manufacturing processes of magnets and explains the assembly of rotor assemblies.

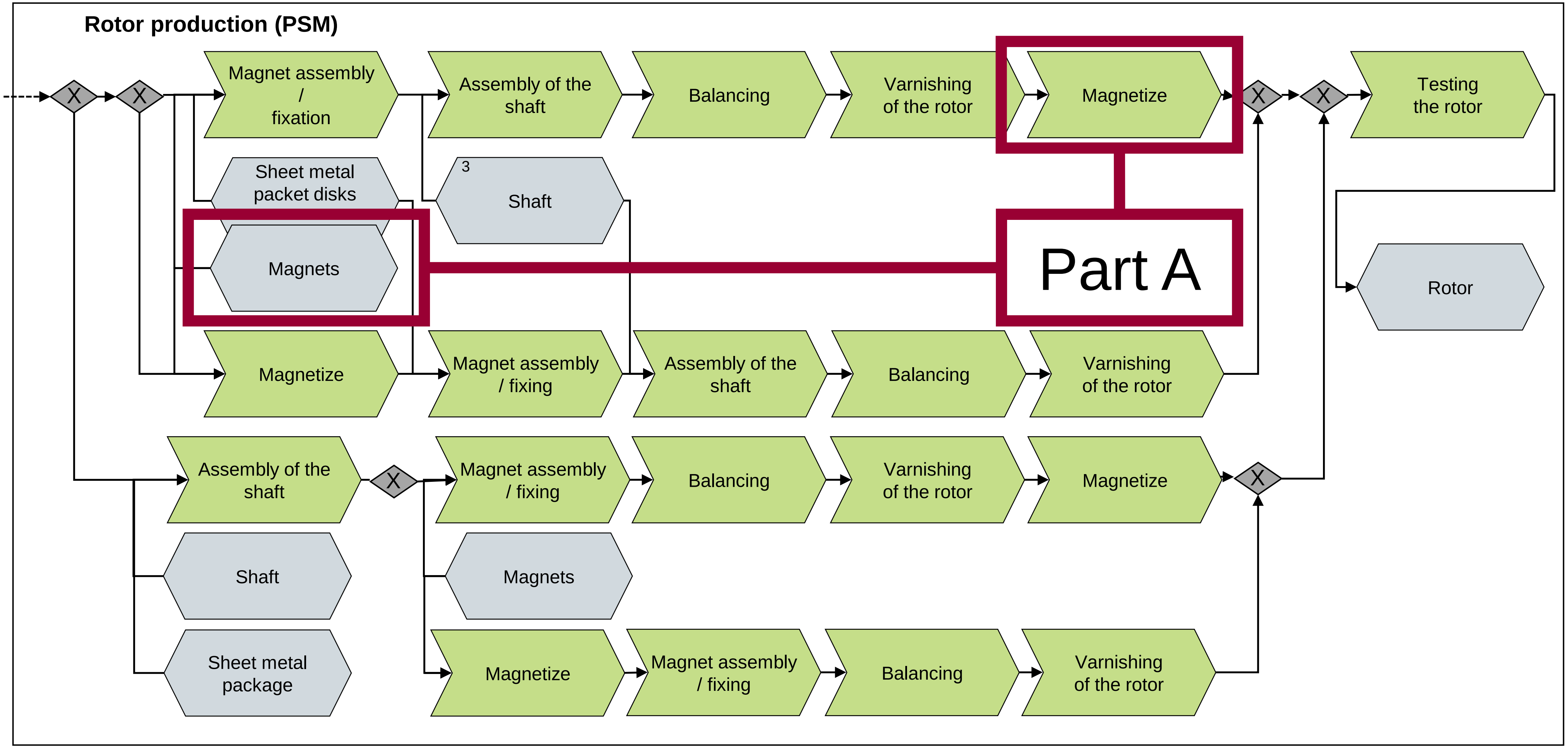
- **Part A: Hard magnetic materials for electrical engineering**
  - Basics of hard magnetic materials
  - Rare earth magnet manufacturing process
  - Magnetization technology for hard magnetic systems
  - Classification and characterization of permanent magnetic materials
  - Material properties of permanent magnets and their microstructure

Graphic: (CC) Flickr.com/Windell Oskay

- Part B: Assembly of permanent magnets
  - Use of permanent magnets in electrical engineering
  - Design of rotor assemblies, rotor topologies
  - Challenges in handling magnets, magnet logistics
  - Assembly and fixing of magnets
  - Fuses for high-speed concepts
  - Qualification of connection technologies

Graphic: FAPS

The lecture unit discusses sections of the manufacturing processes that are used for rotors of a permanent magnet synchronous machine (PSM).



Magnetic materials can be divided into soft magnetic and hard magnetic.

Slide from Lec. 2

Magnetic materials

Sheet metal package

Soft magnetic

Use among others:

- Electric drives (stator / rotor)
- Generators
- Contactors, relays, transformers

Parameters:

- High saturation flux density
- Narrow hysteresis
- Low coercivity

Materials:

FeNi, FeCo, FeSi

Lecture unit 3

Soft magnetic rotor and stator of a reluctance machine

Graphic: ZF Friedrichshafen AG

Hard magnetic

Use among others:

- PM rotors
- Switching magnets
- Holding magnets
- Medical diagnosis
- Sensors

Parameters:

- Wide hysteresis
- High coercivity
- High retentivity

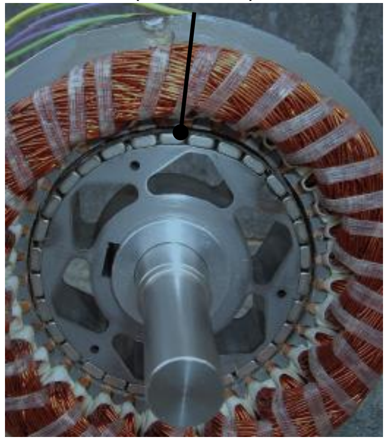
Materials:

Hard ferrites, NdFeB, SmCo, plastic-bonded, AlNiCo

Lecture unit 4

Hard magnetic rotor of a permanent magnet synchronous machine

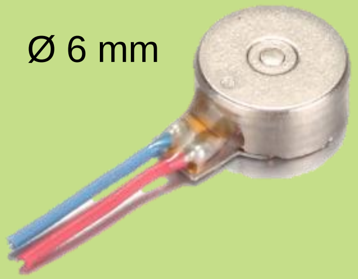
Graphic: Swiss Federal Office of Energy SFOE (Switzerland)



In particular, the use of rare-earth magnets has revolutionized drive technology due to their high power density.

From miniature motors:

- BLDC as vibration motor
- For haptic feedback
- Nominal power: approx. 0.2 W



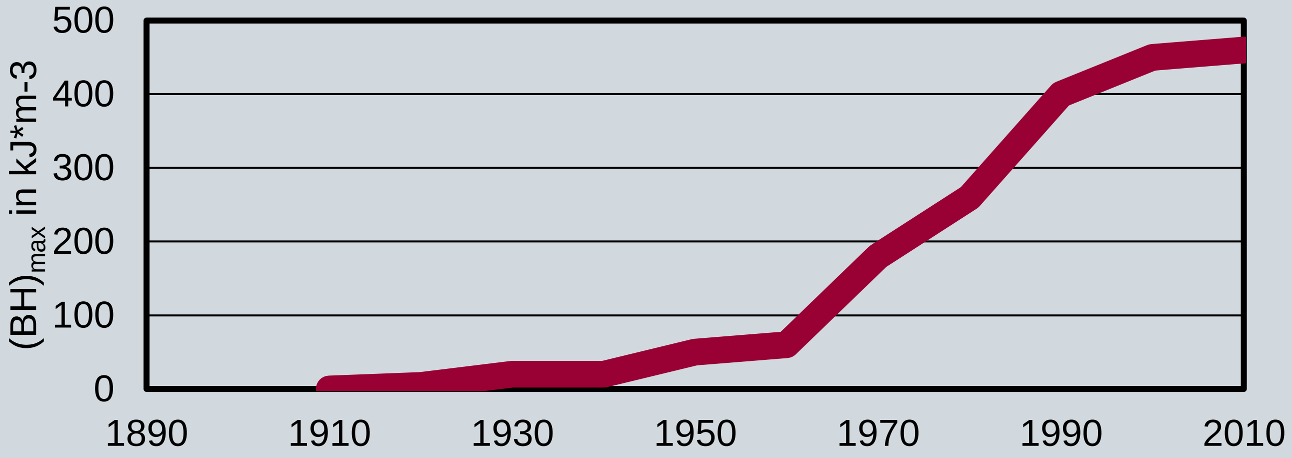
Graphic: Indeedmotors

Up to large engines:

- PM Generator
- For wind turbines
- Power: 8 MW

Graphic: TheSwich.com

Energy product development through the use of rare earth magnets



Peak values of the maximum energy product $(BH)_{max}$ of available magnetic materials in the last 100 years

- Rare earth magnets have significantly increased the efficiency of electric drives
- This was made possible by innovations in materials science

In order to define the right test for permanent magnets, a catalog of requirements is first drawn up that specifies which quality criteria must be met.

Selection criteria for permanent magnetic materials

Selection of the **magnetic material** with regard to logistical, application-specific and economic criteria

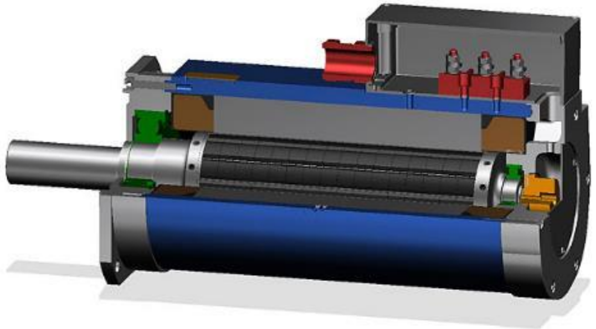
Determination of the **magnet geometry** for reaching the ideal operating point on the hysteresis curve

Checking the **flow distribution** in combination with the stator

Selection of a suitable surface **coating** against corrosion

Maximum operating temperature of the materials against maximum local peak temperature

Quality criteria / test criteria



Graphic: Swiss Federal Office of Energy SFOE (Switzerland)

The saturation flux density is an important characteristic value for soft magnetic materials and indicates the maximum possible magnetization of a material.

Slide from Lec. 2

Magnetic flux density B:

Property of a magnetic field:
$\vec{B} = \mu \cdot \vec{H}$

Magnetic polarization J:

Part of the flux density caused by the magnetic material:
$\vec{J} = \vec{B} - \mu_0 \cdot \vec{H}$

Permeability μ:

Magnetizability of the material in an external magnetic field
$\mu = \mu_0 \cdot \mu_r$

Saturation flux density B_S :

Maximum possible magnetization of a material

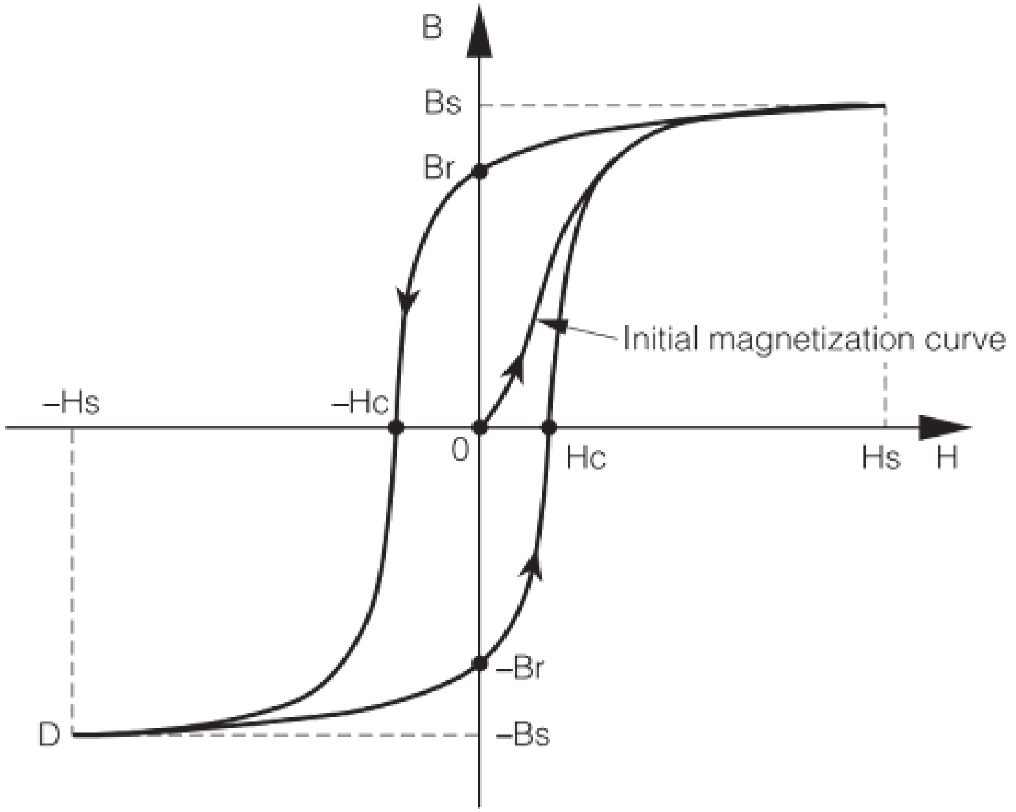
Remanence flux density B_R :

Magnetization that remains after removal of the external field

Coercive field strength H_C :

Field strength for complete demagnetization of a material

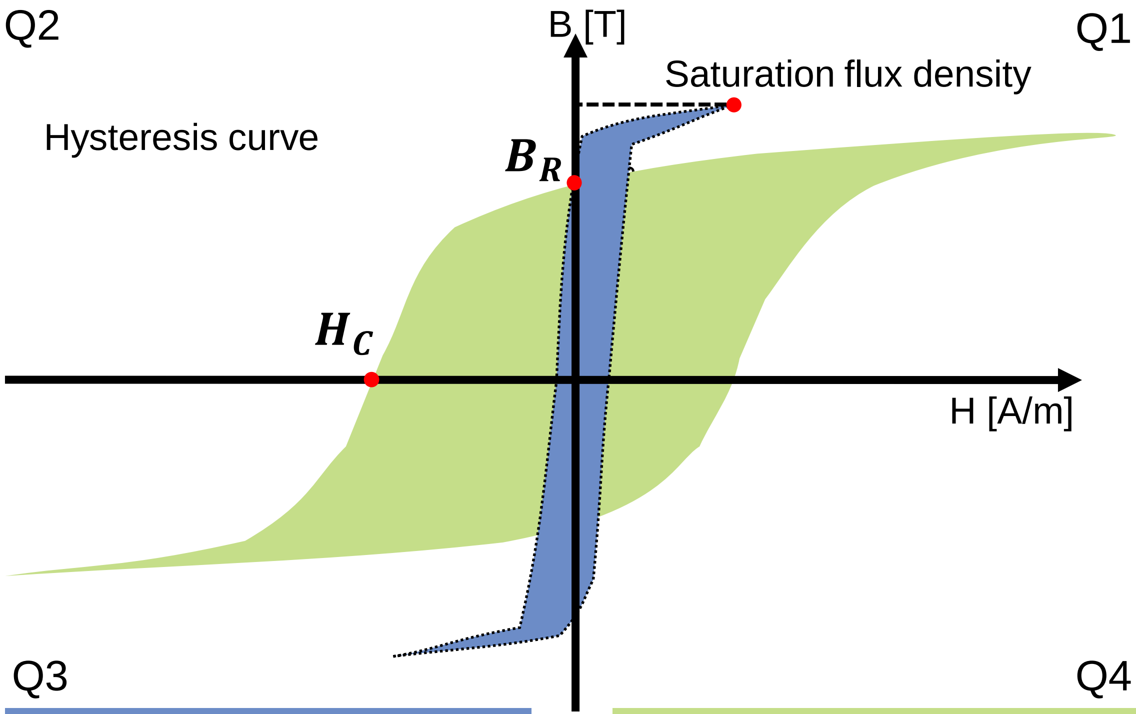
Hysteresis loop of a soft magnetic material



Graphic: owenduffy.net

The magnetization curve provides information about the magnetic behavior of the material during the magnetization process.

Soft magnetic vs. **Hard magnetic**



Soft magnetic

Materials:

- Low coercivity
- High remanence

Hard magnetic

Materials:

- High coercivity
- High remanence

Magnets

Remanence

- Remaining magnetization in a substance that remains after removal of the external magnetic field
- is therefore the value for the strength of a permanent magnet

Coercive field strength

- Stands for the magnetic field strength required to demagnetize ferromagnets, i.e. is equal to zero
- is therefore the stability against demagnetization of a magnet

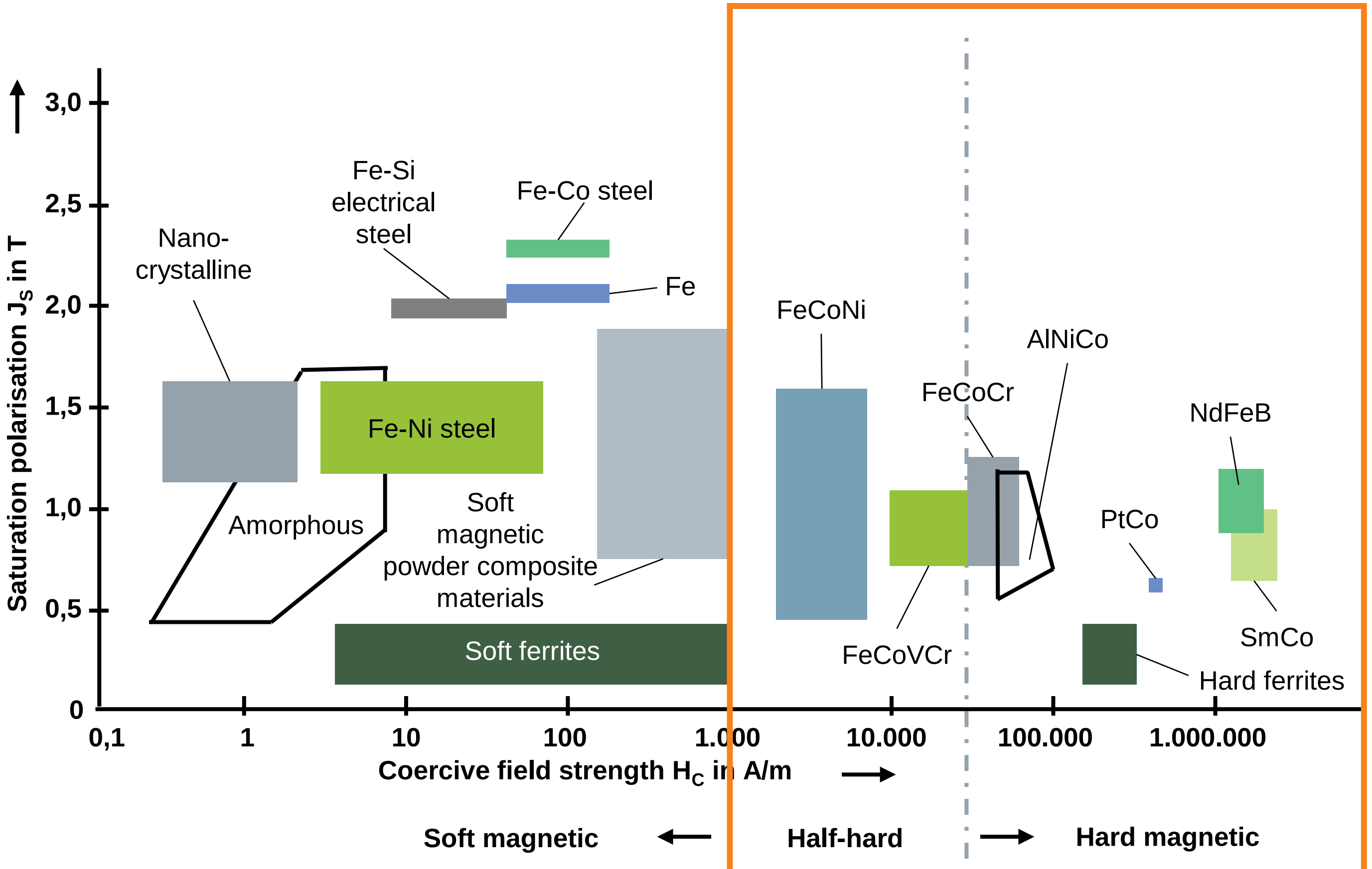
Saturation flux density

Magnetization *M* at which an increase in the external magnetic field strength *H* in the material no longer causes an increase in the magnetization of the material comp. Slide Lec. 2

Soft magnetic materials can be categorized according to saturation polarization and coercivity field strength.

Classification of hard magnetic materials:

- Key parameters:
  - High remanence (retained magnetism) and
  - coercive field strength (Hc) – the field required to demagnetize the material
  - Hard magnetic materials exhibit wide hysteresis loops with high energy loss during magnetization reversal, contrasting sharply with soft materials' narrow loops.
- Material types:
  - Rare-earth magnets (NdFeB, SmCo) with extreme energy products
  - AlNiCo alloys for high-temperature stability
  - Hard ferrites (ceramics) for cost-effective motors/speakers
- Structural features:
  - Powder metallurgy processing ensures dense magnetic alignment
  - Crystalline anisotropy enhances resistance to demagnetization



Graphic: Boll

Magnetic materials are classified by the coercive field strength and saturation induction.

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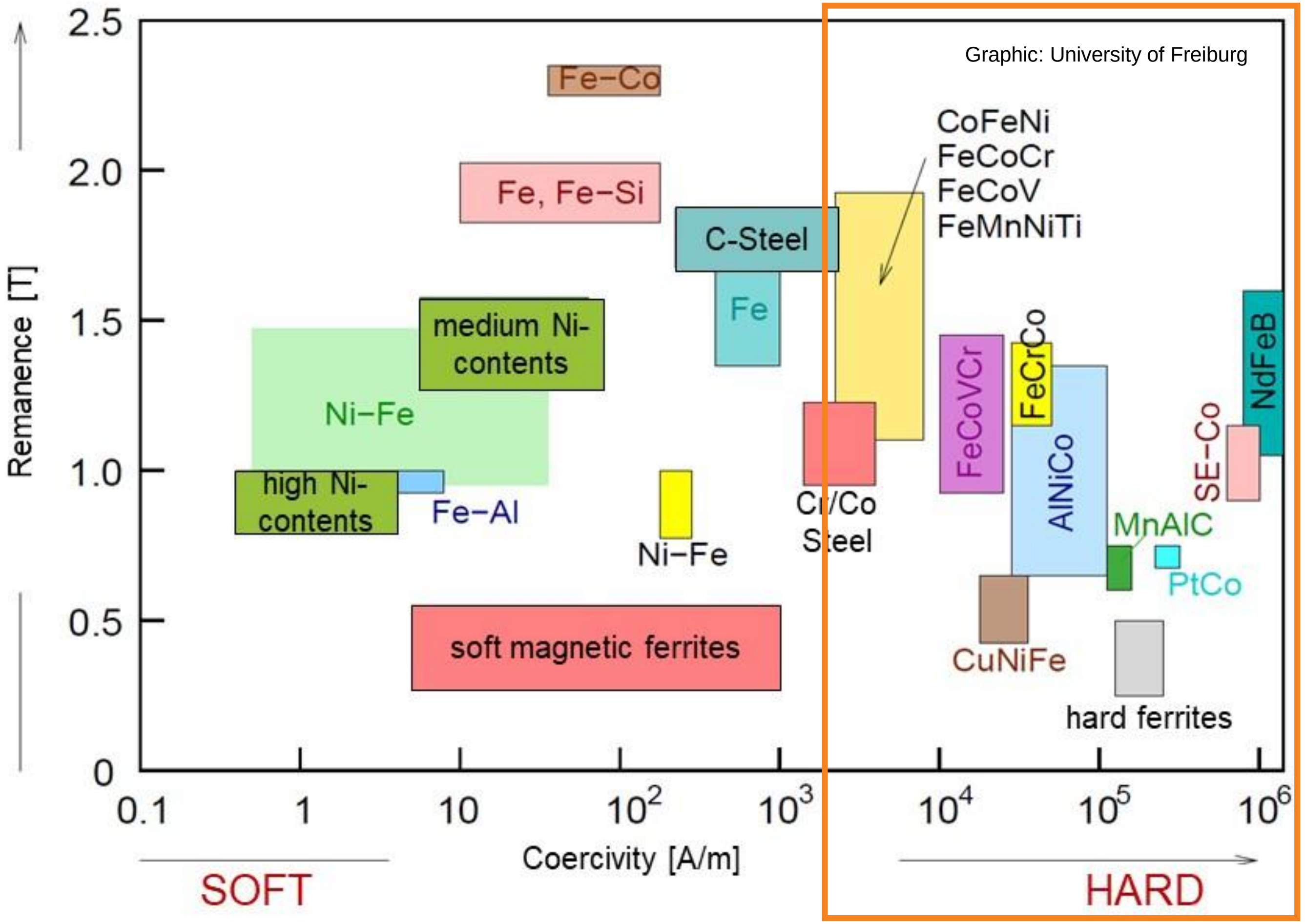
Classification into soft magnetic, semi-hard and hard magnetic materials



Most important hard magnetic materials in electrical engineering

Magnets

- AlNiCo (aluminum-nickel-cobalt)
- Nd / NdFeB (neodymium-iron-boron)
- Ferrite magnets
- SmCo (samarium cobalt)
- NdFeB plastic-bonded
- Hard frits plastic bonded
- SmCo plastic-bonded

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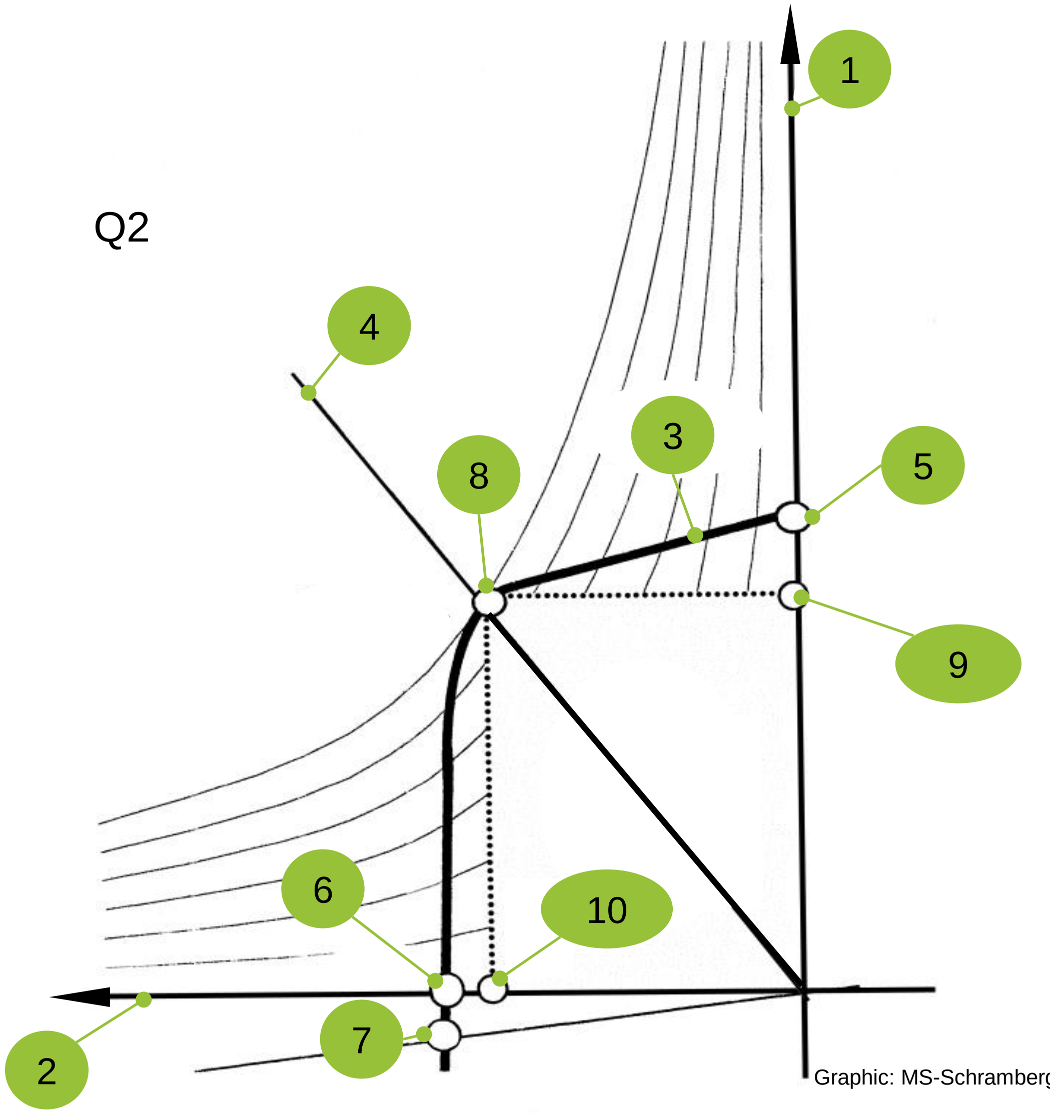
The magnetic energy potential determines the energy available for electromagnetic interaction.



Graphic: MS-Schramberg

Magnets

For the selection of suitable magnets, the 2nd quadrant of the hysteresis curve is important. The cu segment located in the second quadrant of the hysteresis curve is called the demagnetization curve

1. Magnetic flux density (B)
2. Magnetic field strength (H)
3. Polarization curve (J)
4. Working line
5. Remanence (B_R)
6. Flux density - coercivity (H_{CB})
7. Polarization coercivity (H_{CJ})
8. Working point $(BH)_{max}$
9. Working flux density (B_a)
10. Working field strength (H_a)

(B*H)max describes the highest energy density that can be achieved with a material

The determination of the magnetic characteristics of permanent magnets is described in IEC 60404-5.

Magnets

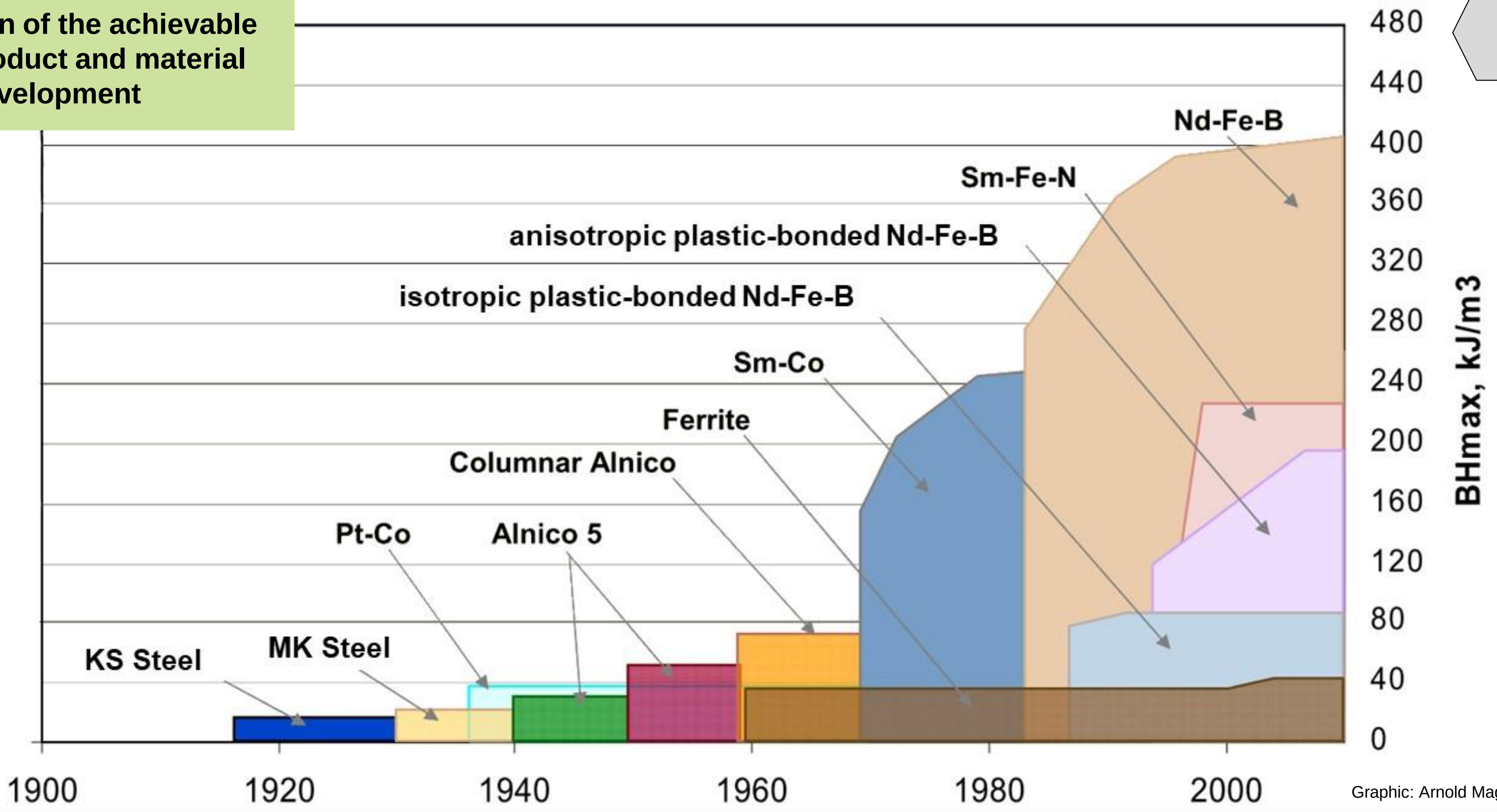
Energy product	$(B*H)_{max}$	J/m³ (type)
Coercive field strength	H_C	A/m (type)
Temperature coefficient	from H_C	% / K
Remanence	B_R	T (typ.)
Temperature coefficient	from B_R	% / K
Magnetization field strength	(not standardized)	A/m
Currie temperature	T_{Currie}	°C
Operating temperature	T_{max}	°C
Density	ρ	g/cm³

Values are given in the temperature range between 20 and 100 °C

For plastic-bonded magnets, the operating temperature is specified for the respective plastic

Rare earth magnets have provided a major step forward in magnet technology, and the increase in achievable energy product continues.

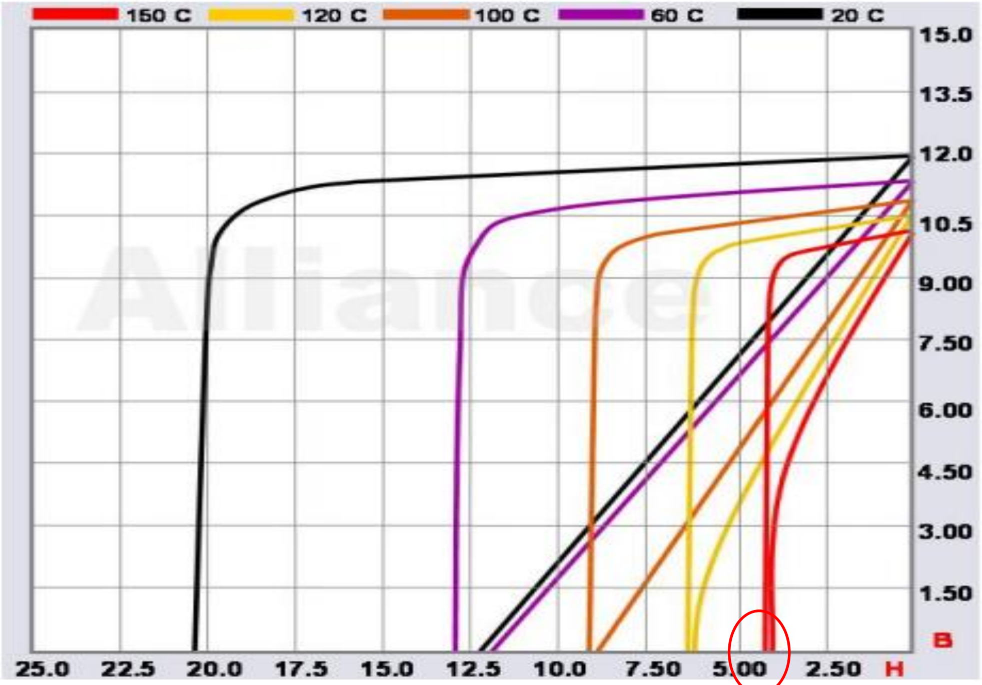
Progression of the achievable energy product and material development



Graphic: Arnold Magnetics

The rare earth magnets SmCo and NdFeB in particular have revolutionized the production of electrical machines with their high demagnetization field strengths.

Demagnetization characteristics for neodymium-iron-boron

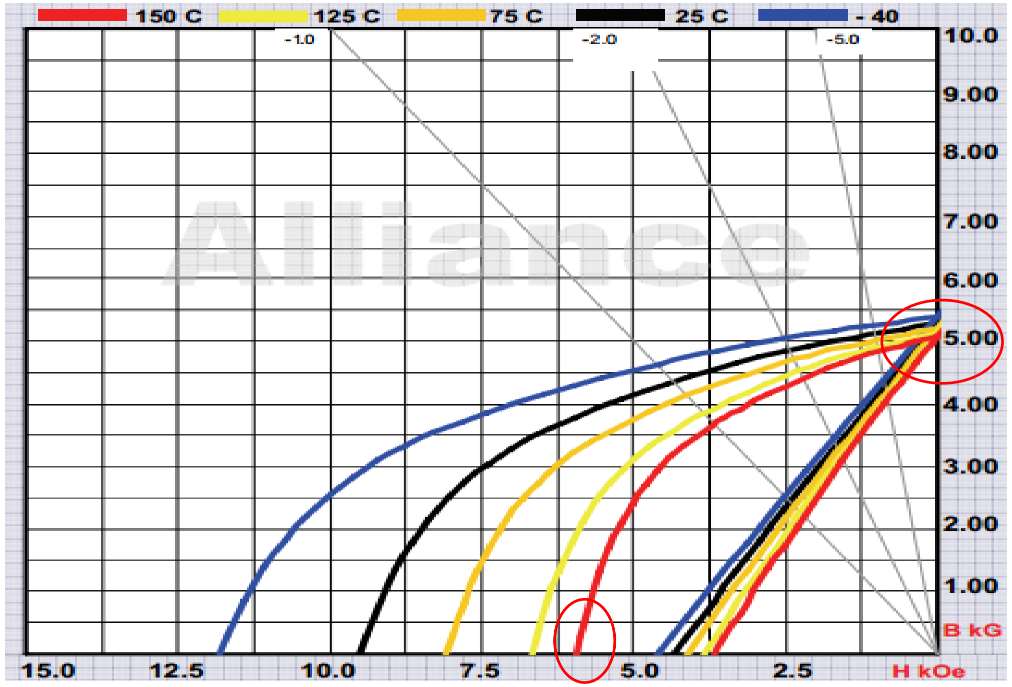


Graphic: Alliance LLC

Advantages:

- High remanence
- High coercivity at low temperatures

Demagnetization characteristics for samarium-cobalt

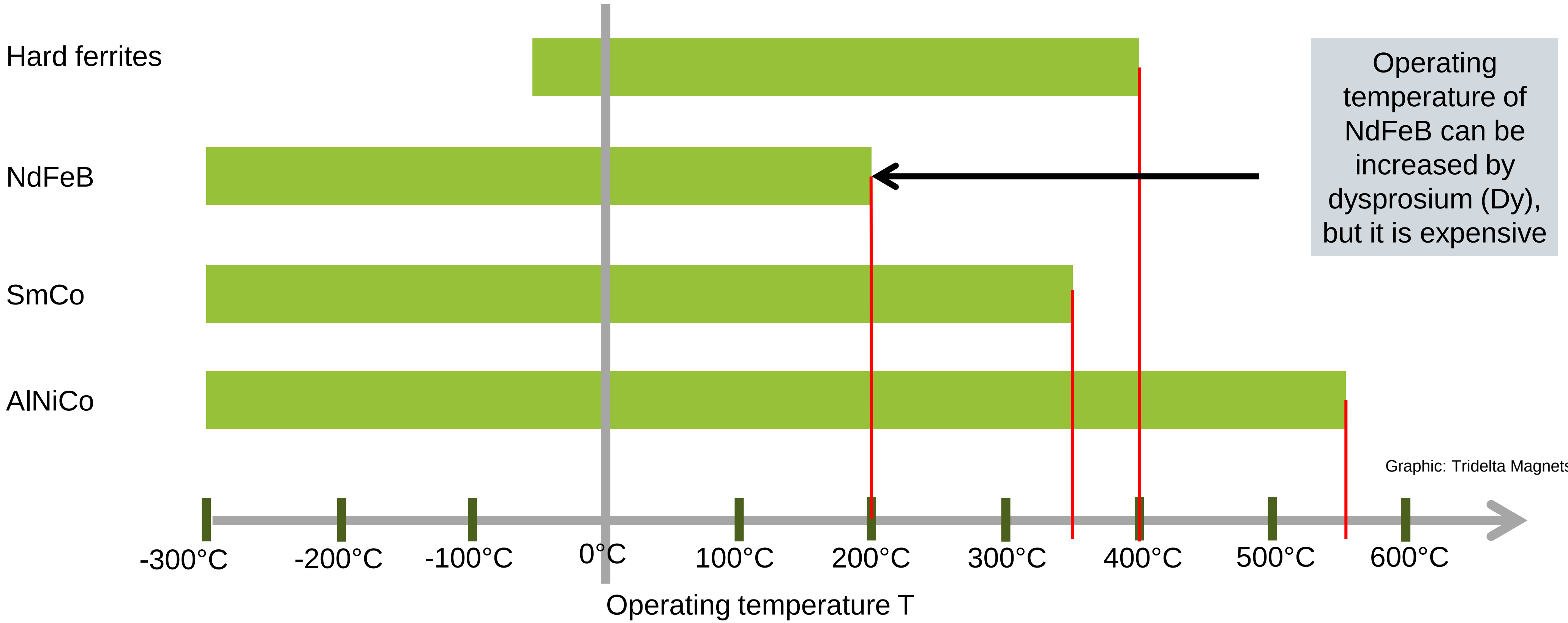


Graphic: Alliance LLC

Advantages:

- Thermally stable
- High coercivity at high temperatures

For high-temperature applications, the choice of rare earth magnets is limited.

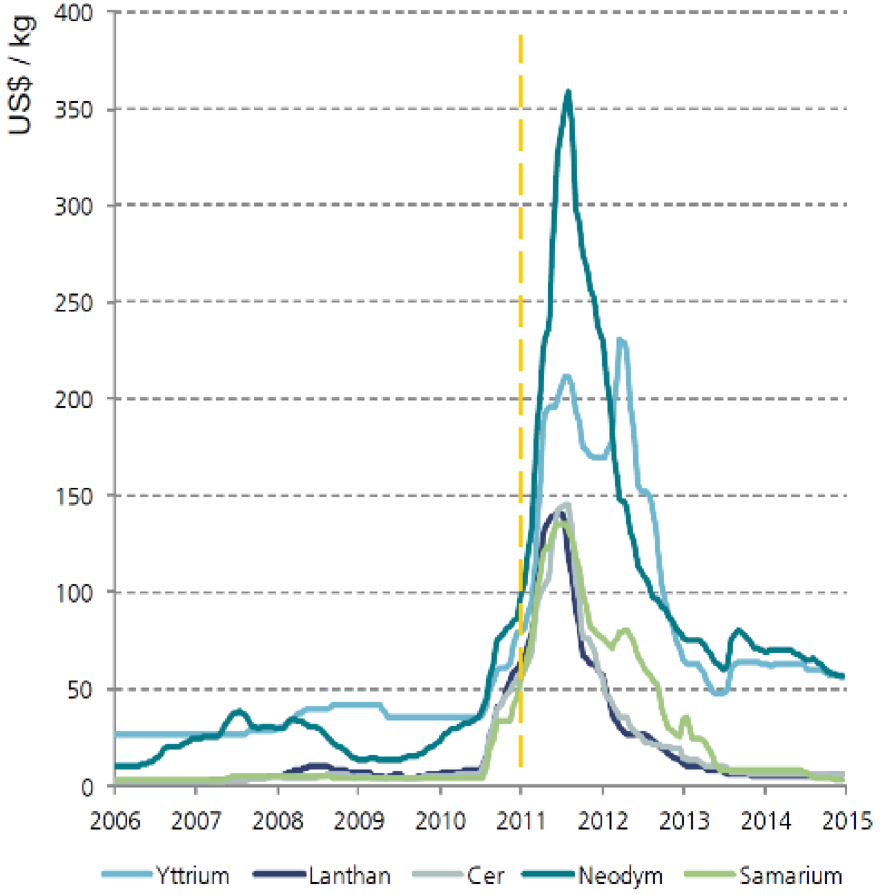


Graphic: Tridelta Magnetsysteme GmbH

At temperatures > 400°C, the use of rare earths is limited.	Limit value can only be achieved with additives

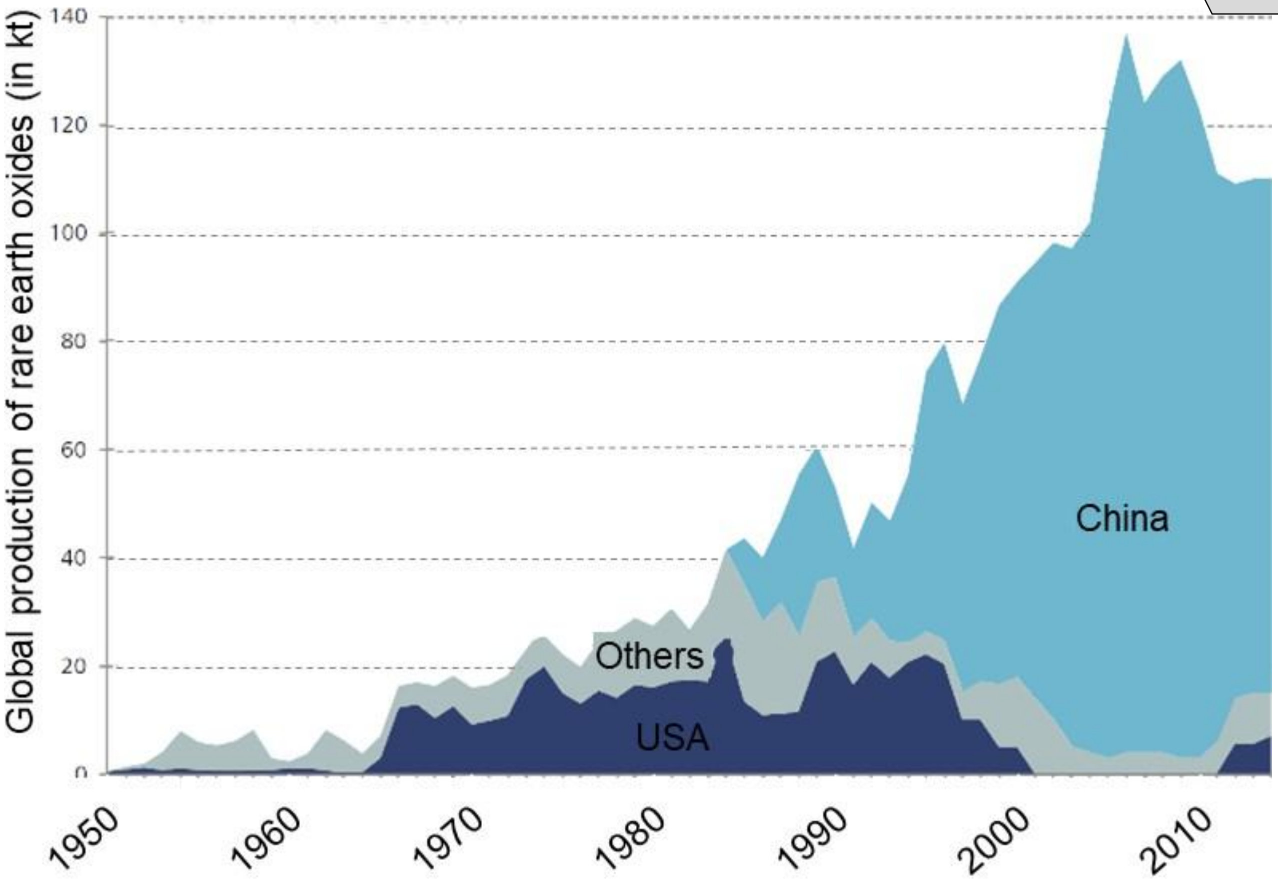
The production of rare earth magnets is increasing, but risks in the supply of raw materials must be taken into account.

Price development of rare earth oxides (prices from China)



Most important rare earth oxide producing countries

Magnets



Source: S. Glöser-Chahoud, A. Kühn

One factor for the efficiency and cost-effectiveness of electric drives is the correct selection of permanent magnetic materials.

Magnetic material	Energy product $(BH)_{max}$ [kJ/m³]	Coercive field strength (H_C) [A/m]	Remanence (B_R) [mT]	Magnetizing field strength (H) [kA/m]	Curie temperature (T_{Curie}) [°C]	Operating temperature (T) [°C]
Hard ferrites	8-25	130-300	200-410	>1000	450	max. 250
SmCo (sintered)	140-195	680-800	900-1100	>3500	720-800	250-300
NdFeB (sintered)	180-280	720-1000	1000-1380	>2000	310-350	120-180
AlNiCo 45/5	36	40-125	1120	200-600	700-850	450-500
Hard ferrites (plastic-bonded and sprayed)	2-18	75-190	110-280	>800	450	120-220 (depending on the plastic)
NdFeB (plastic-bonded and sprayed)	55-80	380-470	580-700	>2800	310	130-140
NdFeB (plastic-bonded and pressed)	30-52	290-490	420-700	>2800	310-350	120-180 (depending on the plastic)

Sources: MS-Schramberg, Maurer Magnetic AG, IBS Magnet, Dr. Hahn GmbH

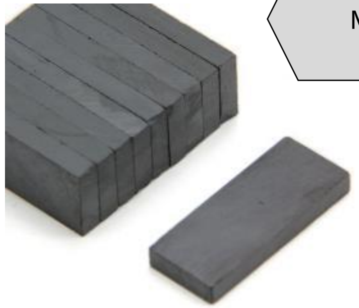
The respective materials must be selected based on the conditions of use.

Magnets

AlNiCo magnets
+ High thermal stability
+ High chemical resistance
- Low coercivity
- Lower remanence than SE
Production: sintering or injection molding

Graphic: magnet4sale

Ferrite magnets
+ Favorable price
+ Good coercivity
+ Oxidation resistant
- Low remanence
Production: pressing and sintering or only dry or wet pressing



Graphic: first4magnets

SmCo magnets
+ High temperature resistance
+ High remanence
+ High coercivity
+ High chemical resistance
- Expensive
- Difficult to rework
Production: Pressing and sintering

Graphic: Euromag

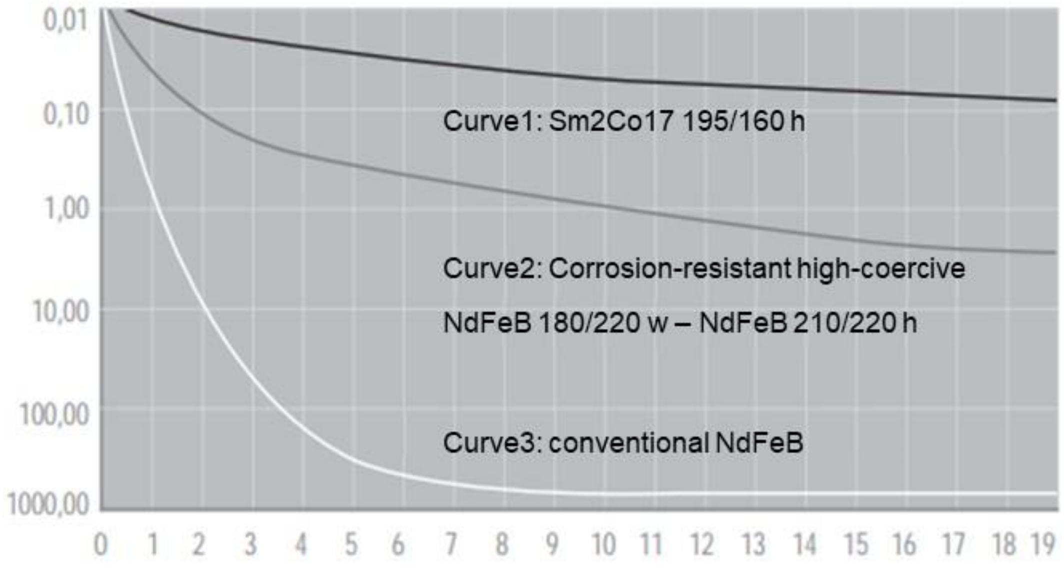
NdFeB magnets
+ High remanence
+ High coercivity
- Low temperature resistance
- Low corrosion resistance
- Difficult to rework
Production: Pressing and sintering

Graphic: Hawell Magnetics

The alloys of rare earth metals in particular are very reactive and and sensitive to oxidation, which makes coating necessary.

Comparison of the corrosion behavior

(Area loss in mg/cm² per day)



Graphic: Tridelta Magnetsysteme GmbH

- NdFeB dissolves completely within a few days (curve 3)
- NdFeB also chemically stabilized by processing (curve 2)
- Sm2Co17 shows only minimal signs of corrosion in the autoclave (curve 1)

Comparison of coatings

Before test 24 hours 72 hou

Neodymium magnet (Without coating) Corrosion resistance: Low

Neodymium magnet (nickel-plated) Corrosion resistance: Very good

Samarium-cobalt magnet (Without coating) Corrosion resistance: Good

Samarium-cobalt magnet (nickel-plated) Corrosion resistance: Very good

According to the tests in the humidity cabinet JIS K 5400 9.2.2: The test temperature is 70°C, while the standard temperature is 50°C. Stains on the cobalt magnet are caused by water droplets.

Source and Graphic: misumi-ec.com

The magnetization curve provides information about the magnetic behavior of the material during the magnetization process.

Slide from Lec. 2

Phases of magnetization in the alternating field:

1: Non-magnetized material

1-2: As the field strength H increases, **domains** with an orientation parallel to the field lines of the external magnetic field **grow:**

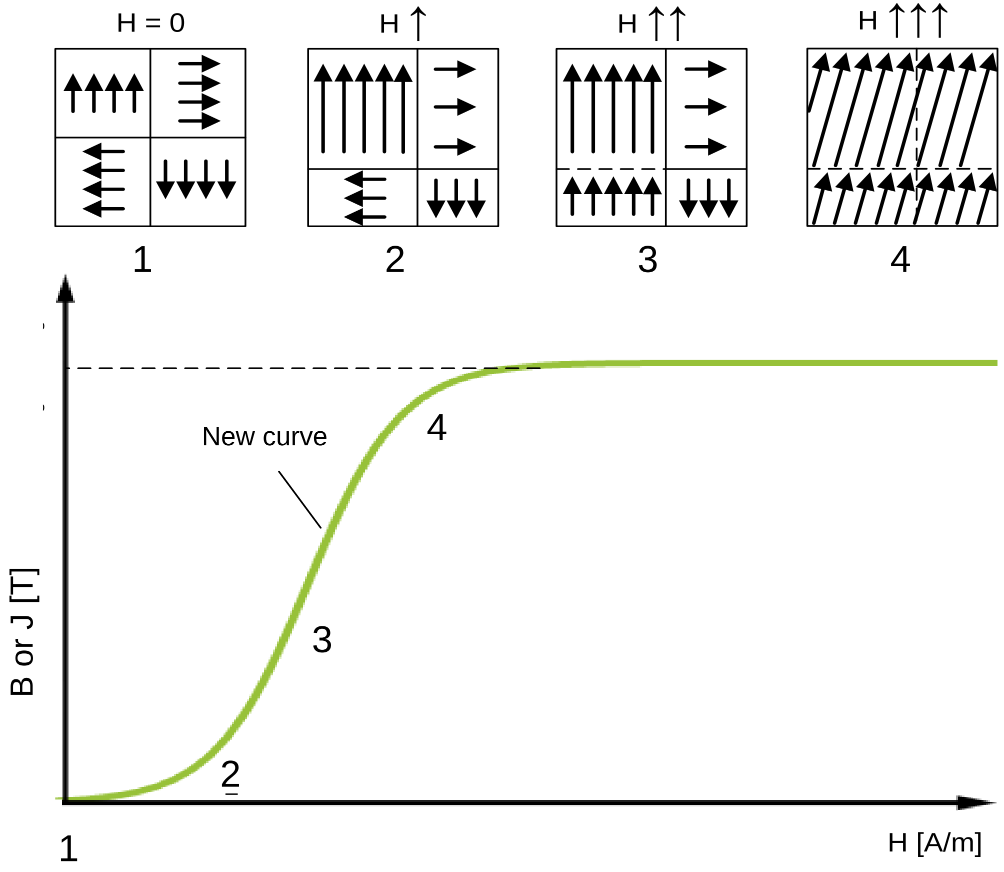
Reversible processes (alignment of the domains (Weiss districts) in the H-field without Bloch wall displacement)

2-3: Irreversible processes in the form of Bloch wall shifts: Enlargement of the domains up to defects → Barkhausen jumps

3-4: Alignment of the domains that are not yet aligned by **rotating the elementary magnetic dipoles**

=> All domains are aligned → Saturation polarization is reached

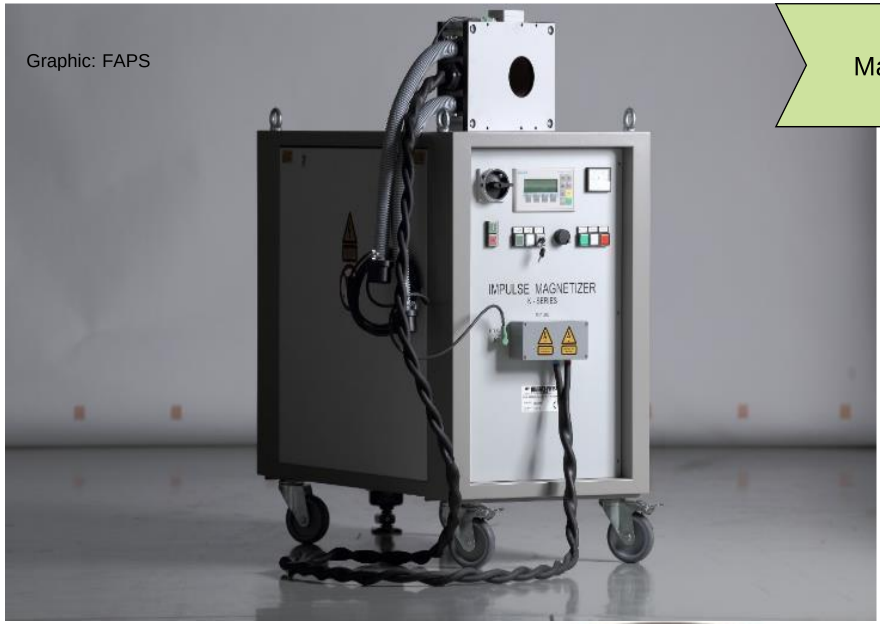
New curve for magnetization process in ferromagnetic material



Graphic: based on Callister et al. 2013

Magnetization using a pulse magnetizer aligns the elementary dipole moments in the material.

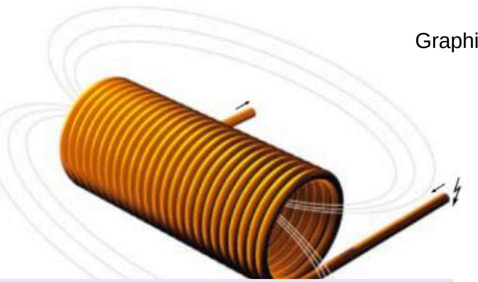
- Isotropic magnets can be magnetized in any direction, as their dipole moments have no preferred direction. Anisotropic magnets, on the other hand, can only be magnetized in their preferred direction, which is determined by their shape.
- Permanent magnets or electromagnets are used for magnetization. They must be able to magnetize the material up to the saturation field strength.
- The most common process is pulse magnetization. Here, energy is first stored in capacitors, which then flows through a wound coil and generates short, strong magnetic fields.
- Depending on the coil type and arrangement, different magnetization directions result.



Graphic: FAPS

Magnetize

Graphics: MS-Schramberg



Magnetizing coil for axial and diametral magnetization

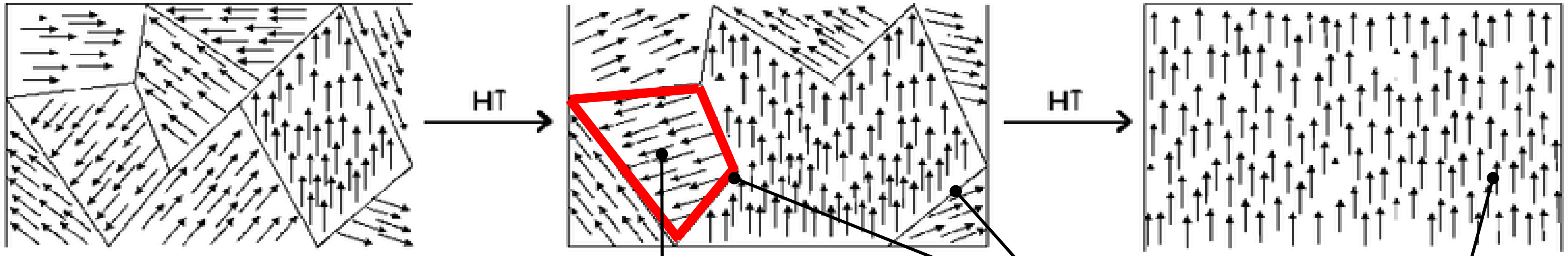
Magnetizing coil for 8-pole magnetization on the inner diameter

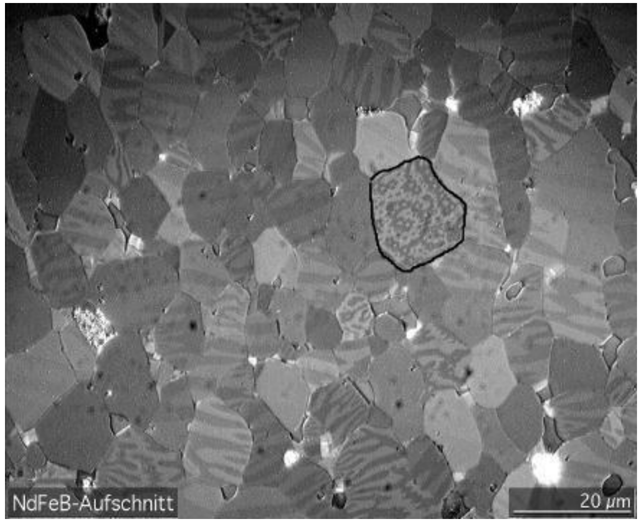
Magnetizing coil for 8-pole magnetization on the circumference

Magnetizing coil for radial magnetization

Magnetic materials consist of domains, which are aligned by an external magnetic field.

Demagnetized	Moving the Bloch walls	Magnetized	Magnetize





Size: 10 µm to 1 mm

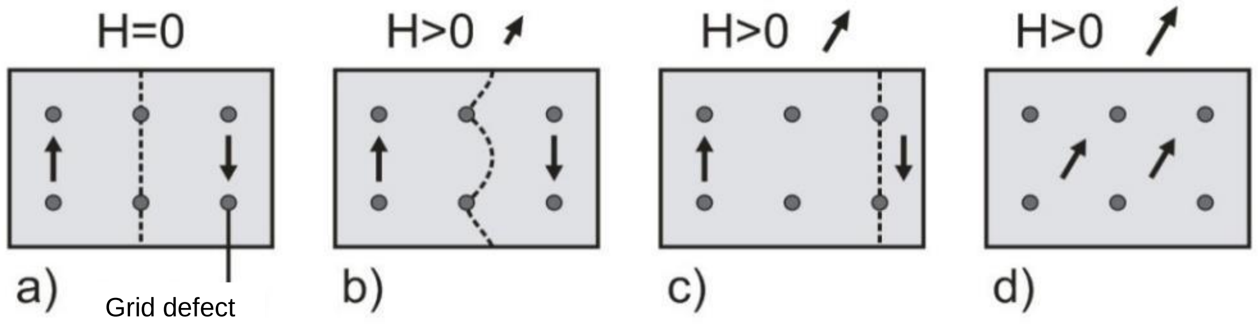
White districts	Bloch walls	Magnetization
Microscopically magnetized domains of ferromagnetic materials	Boundary between the Weiss regions in ferromagnetic materials	Alignment of the electron spin

Graphics: Gaudsmit Magnetic Systemes

The magnetization behaviour of the magnetic material is determined by the Bloch wall mobility.

Magnetize

Bloch wall movement is impeded by grid defects.

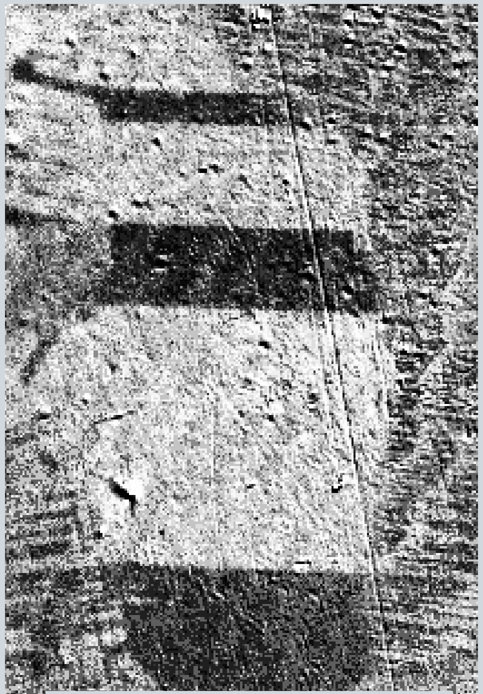


Graphic: Materials in electrical engineering

Increasing field strength H

a) to b)	Reversible wall displacement processes
c)	Irreversible wall displacement processes
c) to d)	Reversible turning processes

The **movement of Bloch walls** can be observed using a Kerr cell microscope. Here in SiFe, under the influence of an external magnetic field.



Graphic: wikipedia.org/wiki/Weiss-Bezirk#

Wall displacement processes occur abruptly, which is why they are also referred to as BARKHAUSEN jumps.

The magnetic force, process and prices depend on whether the magnetic powder is anisotropic or isotropic.

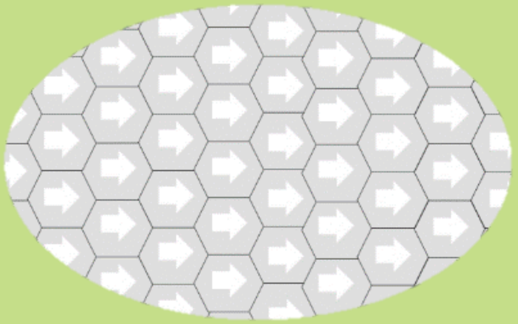
FAPS

Magnets

Elementary magnets are aligned in the same way.

- Preferred direction of the base material
- Strong magnetization in preferred direction

Anisotropic magnets

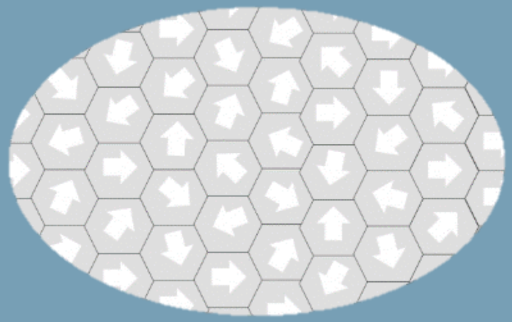


- \+ High magnetic forces / remanence in preferred direction
- − Difficult production
- − Expensive
- − Influence of angle errors on maximum force
- − Straightening field during shaping

Elementary magnets are distributed differently.

- Elementary magnets that were already aligned in the direction of magnetization are magnetized
- Weaker than anisotropic magnets

Isotropic magnets



Graphics: MS-Schramberg

- \+ Cheaper to produce
- \+ Same magnetic property in all directions
- − Moderate, equal magnetic force/resistance in all directions

Plastic-bonded rare earth magnets have established themselves on the market due to their low manufacturing costs and shape flexibility.

Magnets

The mechanical properties of permanent magnets must be differentiated with regard to their manufacturing processes.

Sintered magnets

- Brittle
- High tensile strength
- Poor thermal resistance
- Poor chemical resistance
- Comparatively expensive

Hard ferrites

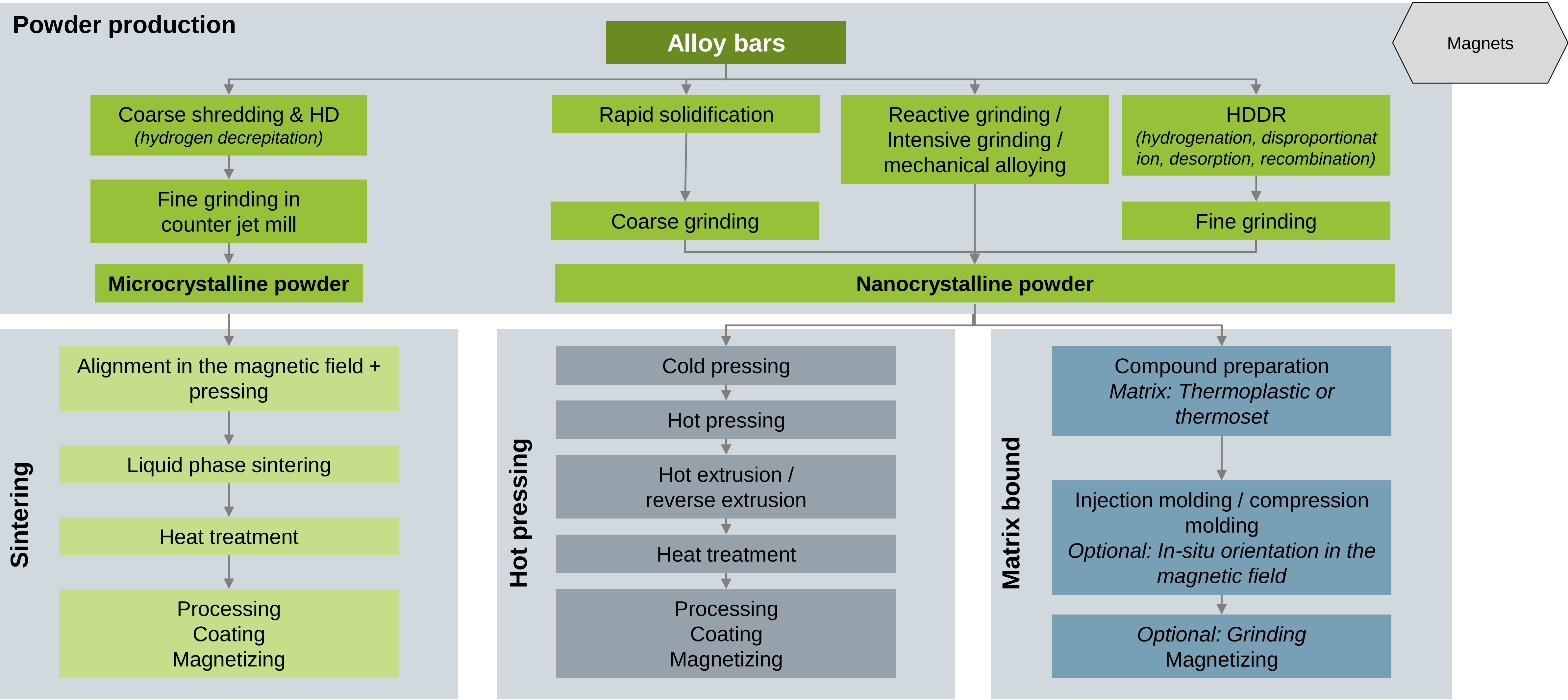
- Brittle
- Average tensile strength
- Medium thermal resistance
- Good chemical resistance
- Comparatively low cost

Plastic-bonded magnets

- Ductile
- Poorer tensile strength
- Poor thermal resistance
- Good chemical resistance
- Many shapes can be realized

Graphics: MS-Schramberg

Rare earth magnets can be produced in three ways, with significant differences in the starting powders.



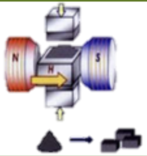
To produce anisotropic magnets, single-crystal powder is aligned in the preferred direction by pressing in a magnetic field.

Magnets

Melting

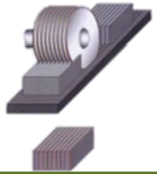
Breaking and HD

Grinding

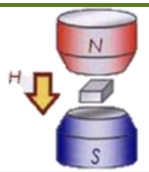


Pressing in a magnetic field

Sintering

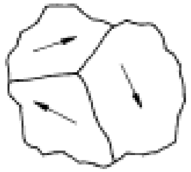


Finishing

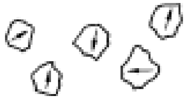


Magnetize

Source: Shin Etzu

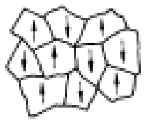


Domains are stochastically distributed (no resulting magnetization)



Monocrystalline powder due to hydrogen embrittlement (HD)

During pressing in the magnetic field, the powder particles are aligned in the preferred (light) direction



The green parts are liquid phase sintered to form a mechanically stable body and are non-magnetic after the process

Source: Hagen Symposium: Powder Metallurgy

All elementary magnets are aligned during magnetization

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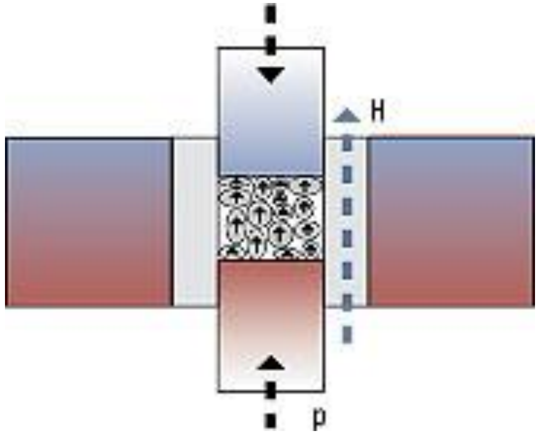
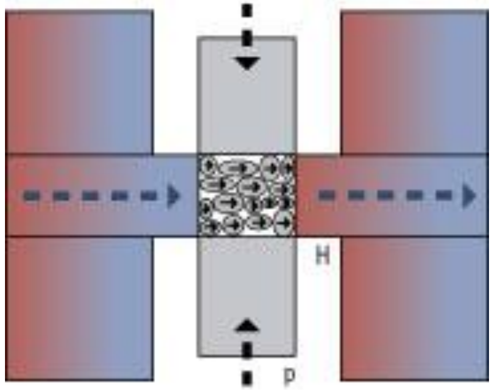
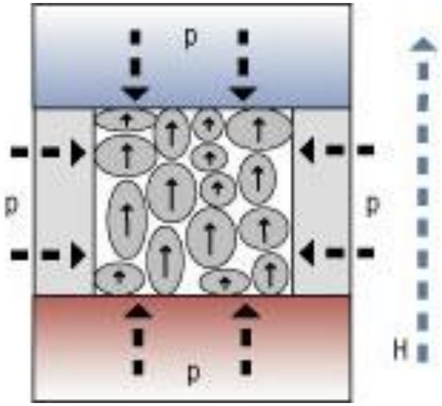
The "green compact" is produced by compacting powder in presses.

In the production of neodymium magnets, the monocrystalline powder is pressed into dies at pressures of 100 to 1000 MPa and under the simultaneous influence of a strong magnetic field, which aligns the powder particles crystallographically.

Magnets

Axial field presses	Cross-field presses	Isostatic pressing
		

Axial field presses

- Most economical process
- Worst magnetic values
- Many shapes can be realized

Cross-field presses

- Near-net-shape pressing, therefore hardly any reworking necessary
- Many shapes can be realized

Isostatic pressing

- Best magnetic values
- Large cubes/cylinders can be realized
- Rework necessary

Graphic: Vacuumschmelze

FAPS

Lecture unit 4 introduces the properties and manufacturing processes of magnets and explains the assembly of rotor assemblies.

Part A

Hard magnetic materials for electrical engineering

- Basics of hard magnetic materials
- Rare earth magnet manufacturing process
- Magnetization technology for hard magnetic systems
- Classification and characterization of permanent magnetic materials
- Material properties of permanent magnets and their microstructure

You learned the following in part A:

- Hysteresis curve of magnetic materials with descriptive characteristic values, in particular the description of the working range of permanent magnets: H_C, B_R, etc.
- Differentiation between soft and hard magnetic materials based on key figures and qualitative properties
- The most important magnetic materials in electrical engineering with value ranges of their material properties and applications
- Theory of the magnetization process in the material
- Manufacturing process for permanent magnets: powder production and pressing process

Lecture unit 4 introduces the properties and manufacturing processes of magnets and explains the assembly of rotor assemblies.

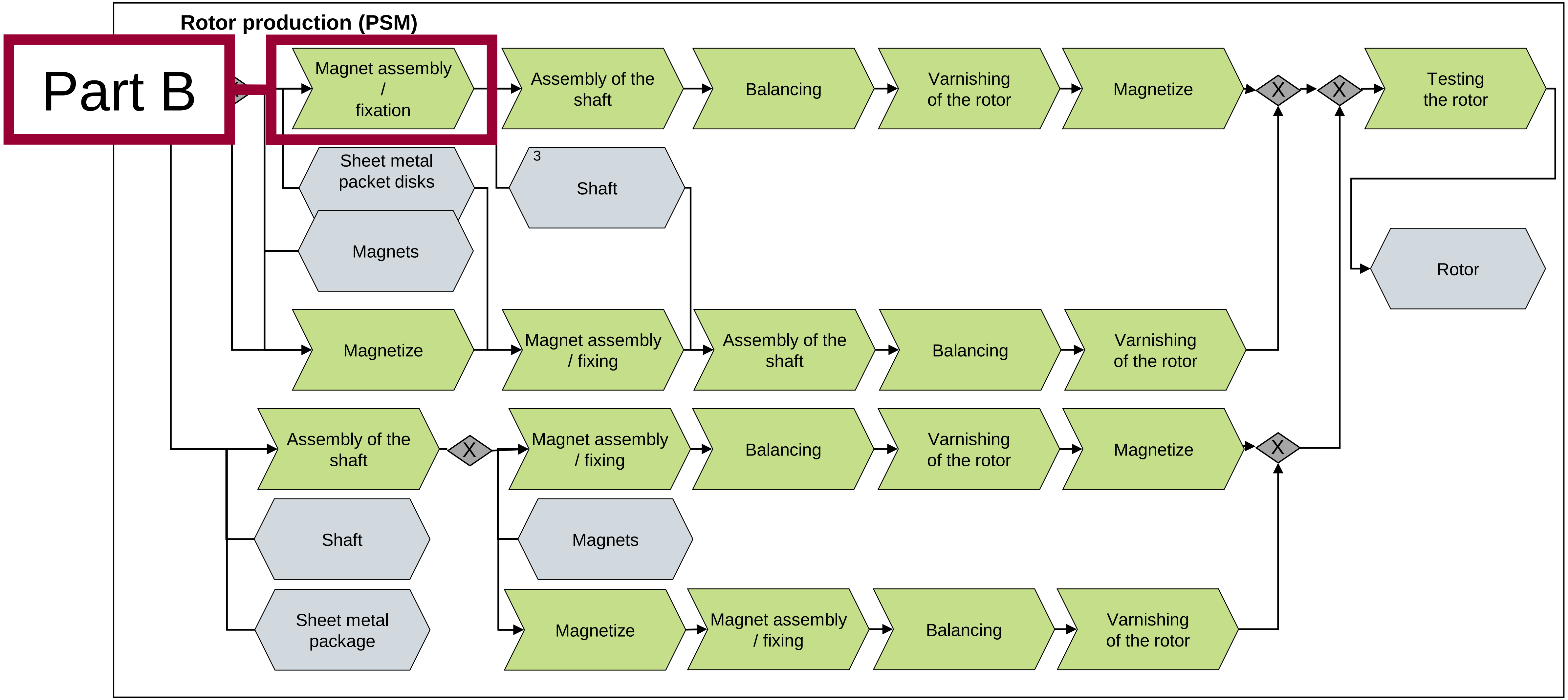
- Part A: Hard magnetic materials for electrical engineering
  - Basics of hard magnetic materials
  - Rare earth magnet manufacturing process
  - Magnetization technology for hard magnetic systems
  - Classification and characterization of permanent magnetic materials
  - Material properties of permanent magnets and their microstructure

- **Part B: Assembly of permanent magnets**
  - Use of permanent magnets in electrical engineering
  - Design of rotor assemblies, rotor topologies
  - Challenges in handling magnets, magnet logistics
  - Assembly and fixing of magnets
  - Fuses for high-speed concepts
  - Qualification of connection technologies

Graphic: (CC) Flickr.com/Windell Oskay

Graphic: FAPS

The lecture unit discusses sections of the manufacturing processes that are used for rotors of a permanent magnet synchronous machine (PSM).



The assembly of high-coercivity magnetic materials complicates the production of rotors for powerful electric motors.

Electric motor process chain

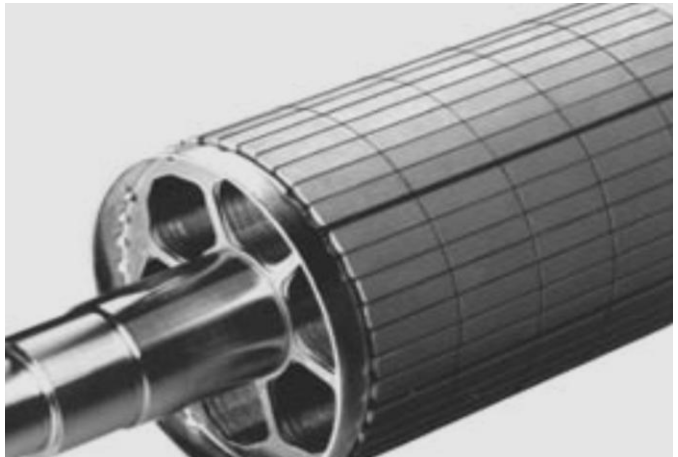
- Stator assembly **60%**
- Rotor assembly **20%**
- Final assembly motor **20%**
- **Electric motor**

Rotor process chain

Sheet metal packaging	5%
Shaft-sheet metal package	5%
Magnet assembly	**80%**
Fixing magnets	
Hardening	
Balancing	5%
Magnetize	5%
Ready-to-install rotor	

- High proportion of manual work content in magnet assembly
- Required production flexibility is realized through cost-intensive personnel work
- Placement and magnetization strategies strongly dependent on sales scenarios

Rotor production PSM



Graphic: vacuumschmelze.de

In the field of rotary motor concepts, there are a number of different rotor designs with specific areas of application.

Conventional rotor design

Rotor in the Tesla Model 3
Graphic: Dan Colestock / quora / Bloomberg Technology

Rotor

Rotor design	Properties
Cage rotor	+ robust motor + low mass moment of inertia + high dynamics, high speeds
Torquerotor / Slow runner	+ low speeds, high torque + Current overload capacity + High continuous torques + Use as direct drive
Hollow runner / Bell runner	+ high efficiency + small mass moment of inertia + higher speeds - Heat dissipation problematic
Disk runner	+ small mass moment of inertia + wide speed control range + Good positioning accuracy ~ Torque: 0.5 to 20 Nm - Overload sensitive

ASM rotor of a spindle motor
Graphic: Heise

Graphic: parker.com

Graphic: ATZ Extra (2014)

Rotor (left) of an axial flux machine
Graphic: H.J. Kierstead

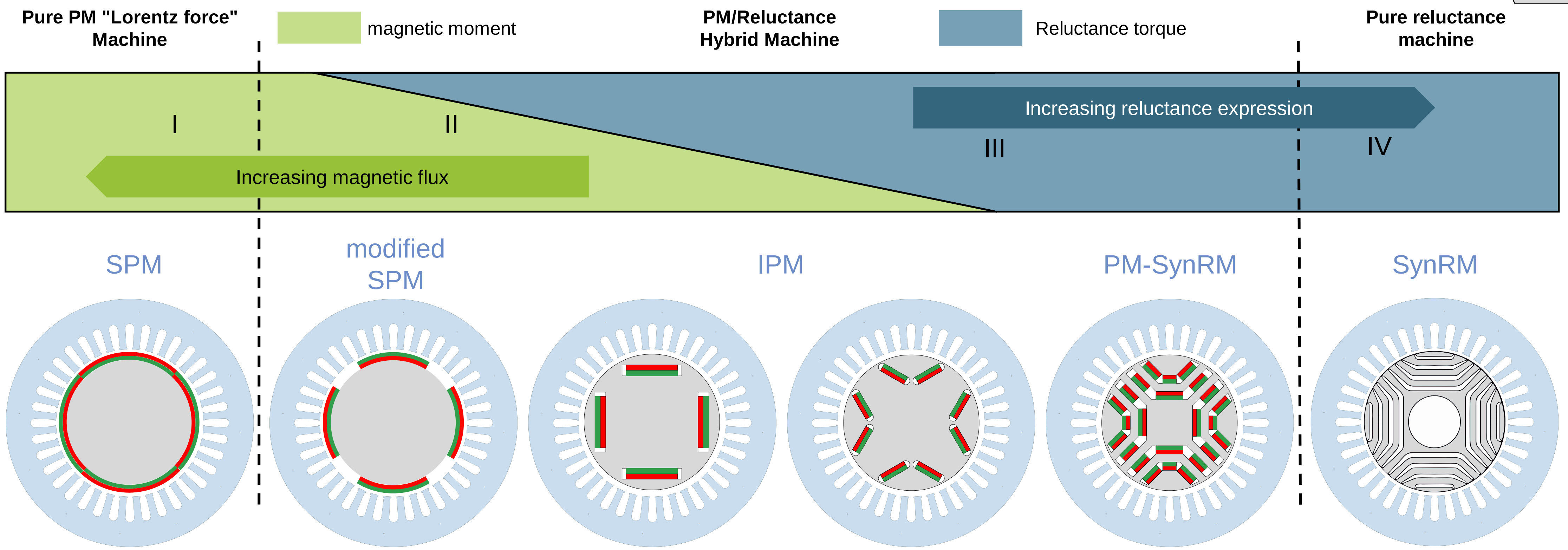
Source for the methodology:
Prof. Dr.-Ing. Eckart Uhlmann, iWF, TU Berlin

Slide from Lec. 2

FAPS

A hybrid of synchronous and reluctance machine can be constructed by selecting the appropriate magnet arrangement and sheet metal cutting geometry.

Rotor



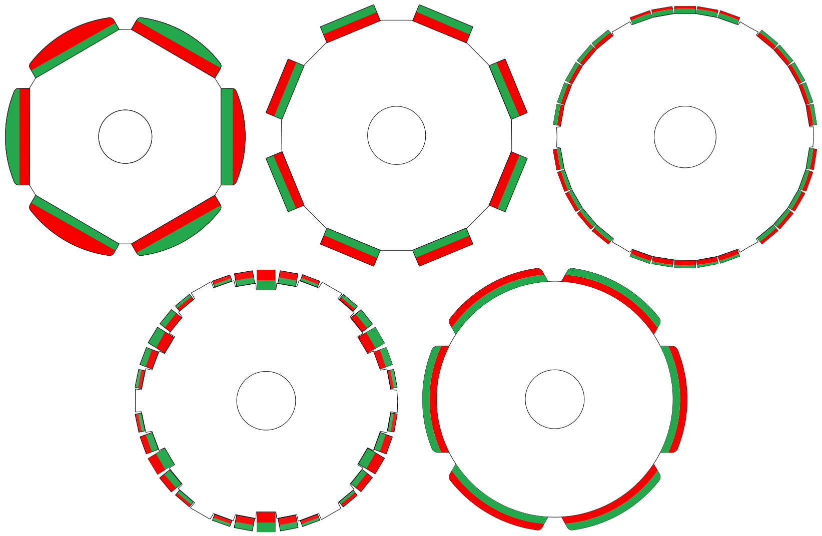
Based on:
Morimoto (IEEE 2007)

The magnets of the permanent-magnet synchronous machine can be mounted on the rotor surface or inserted into internal cavities in the rotor.

Slide from Lec. 2

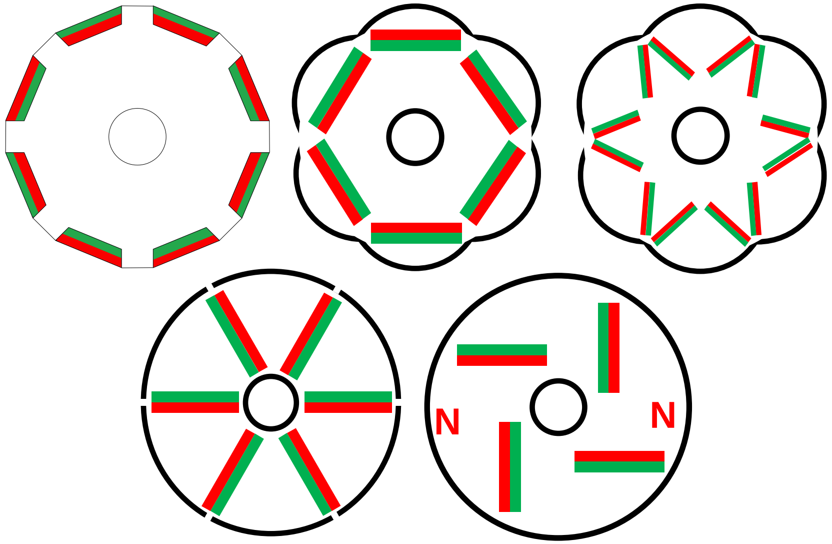
Surface Mounted Permanent Magnets - SPM

- Magnets are glued to the surface of the rotor sheet package
- To maintain rotor mechanical stability under high centripetal forces, a non-magnetic retaining sleeve is required for internal rotors
- SPMs can be dismantled and recycled more easily



Interior Permanent Magnet - IPM

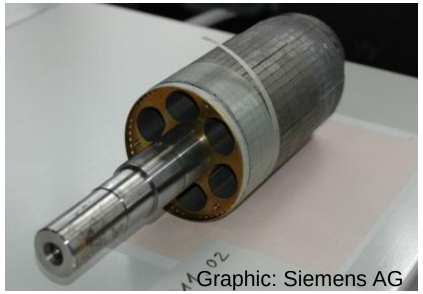
- No gripper required for assembly, so it can be easily automated (e.g. by means of a joining module)
- Depending on the design, high assembly forces must be taken into account
- Air gap can be made smaller than SPM
- No retaining sleeve required
- Dismantling of the magnets is more difficult



The rotor converts a magnetic moment into a mechanical torque and transfers power to the shaft.

Slide from Lec. 1

Assembly	Components	Physical function
Rotor	Rotor shaft	Mechanical transfer of torque Ensure mechanical stability of the lamellar stack
	Rotor sheet metal package	Guide the magnetic field with minimal losses In IPM-Motors: Protect magnets from mechanical damage
	Permanent magnets	Create magnetic poles required to generate mechanical torque
	Taping	Mechanically secure the magnets Protect the magnets from mechanical damage
	End plates	Correction of imbalances Stabilize the lamellar stack by creating axial compression
	Insulating resin/ Glue	Protect magnets from external influences (esp. corrosion) Maintain magnets in fixed position (esp. high rotation speed)

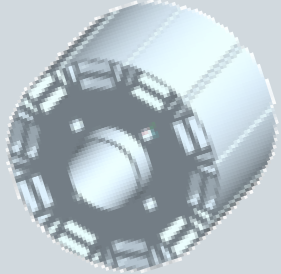

Graphic: Siemens AG

The large-scale production of permanent magnet synchronous rotors requires efficient processes and testing technologies.

Rotor production PSM

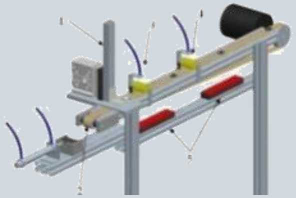
Mounting shaft / sheet metal package

Alternative joining processes for shaft / sheet metal package



Logistics (magnet, rotor)

- Magazine provision and separation

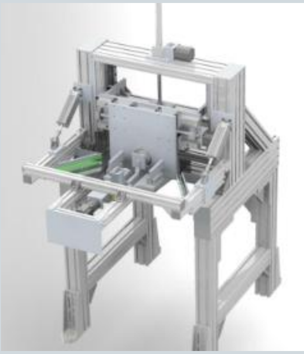


- Workpiece carrier concepts for rotors

Magnet assembly

- Gripper technologies for SPM

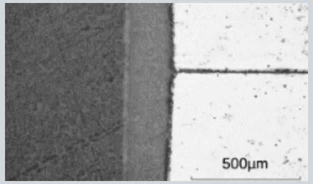
- Slide-in tools for IPM



Magnetic fixation

- Gluing (SPM)

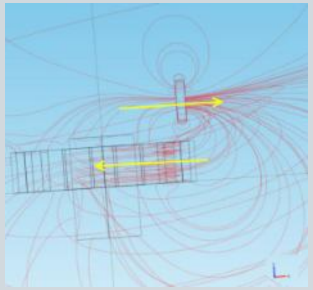
- Bonding (IPM)



- Caulking (IPM)

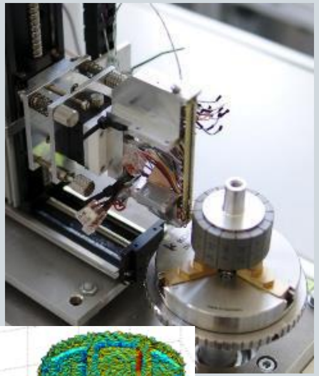
Magnetization

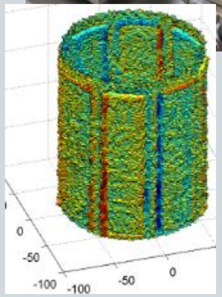
Integration of magnetization in the assembly process



Magnet measurement technology

In-line measurement strategies for checking geometric and magnetic parameters



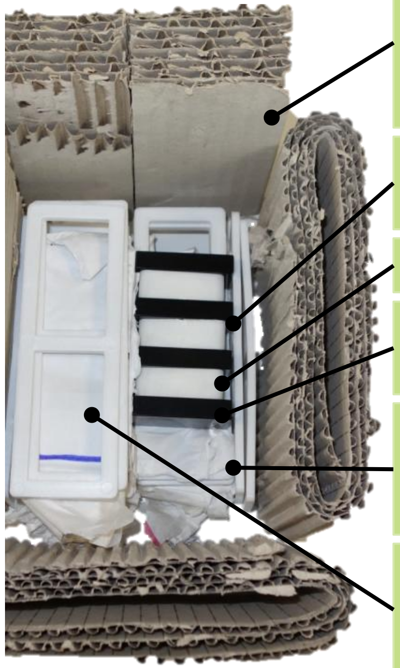


In addition to assembly, magnetized magnets also require special consideration during transportation and separation.

Intralogistics and handling:

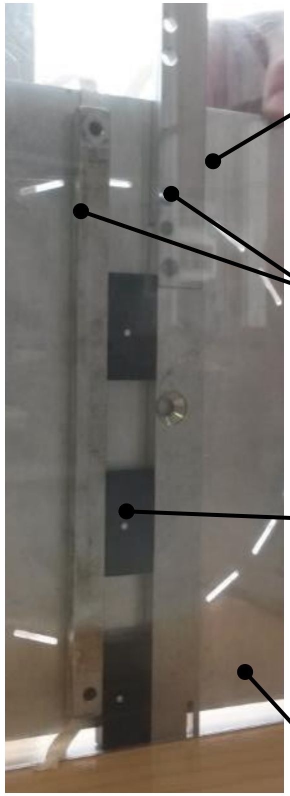
Typical examples for packaging and feeding magnets

Typical packaging of active magnets on delivery



- Paperboard for shock protection and to keep magnetically sensitive objects at a distance
- Double plastic frames to insulate the magnetic stacks
- Plastic spacers
- NeFeB magnet with black epoxy resin coating
- Paper envelope as protection against metal dust and shavings
- Marking of the magnetic alignment. Here: North pole at the bottom

Demonstration of magazine capacity when using magnetized magnets



- Aluminum plate
- Aluminum guides
- Magnets aligned in the same direction
- Acrylic glass pane

General aspects of handling and assembling active solenoids:

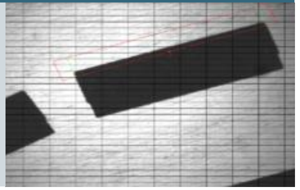
Rotor production PSM

- Hard magnetic materials are fragile: Shocks must be avoided and precise guidance of the magnetic bodies is required to prevent tilting
- Damage to the magnet surface can result in significant degradation of magnetic properties
- Active magnets should be handled with care: Removal from transport packaging, separation of magnets, storage in magazines
- Magnets must be guided in all degrees of freedom
- Observe polarity and orientation during assembly
- Sufficient rigidity of equipment to safely absorb forces
- Devices must generally be made of non-magnetic materials

The handling and assembly of already magnetized magnets is a focus of research at the institute FAPS.

IPM
(Internal Permanent Magnet)

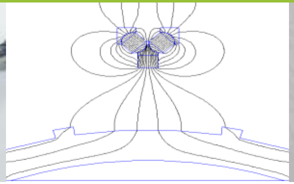
Optical detection of the cavities and positioning of the insert mold



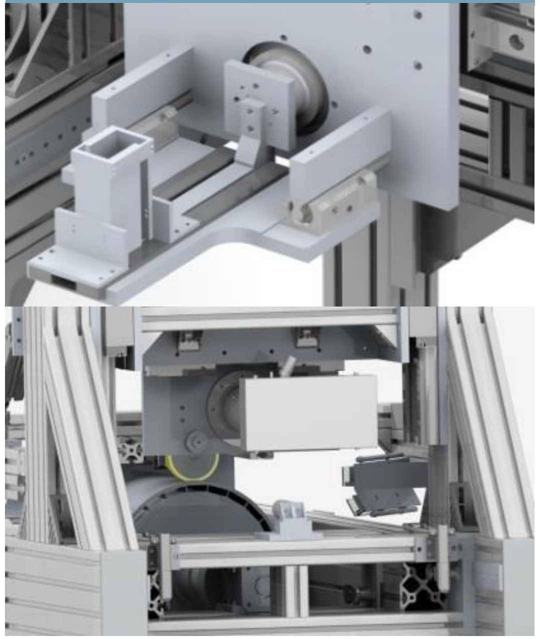
SPM
(Surface-Mounted Permanent Magnet)

Magnet assembly

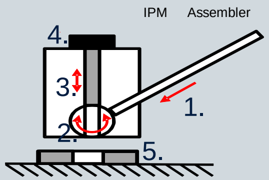
Magnetic gripper



Horizontal insertion

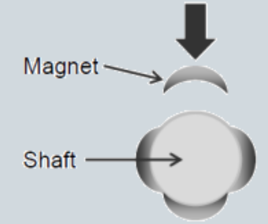


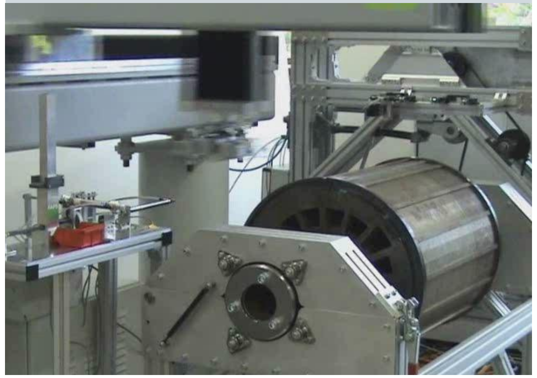
Vertical insertion



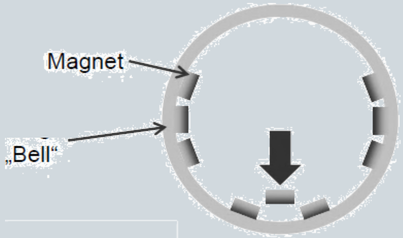
1. Magnet
2. Loading unit
3. Push unit
4. Gripper holder
5. Workpiece

Internal rotor





Outer rotor



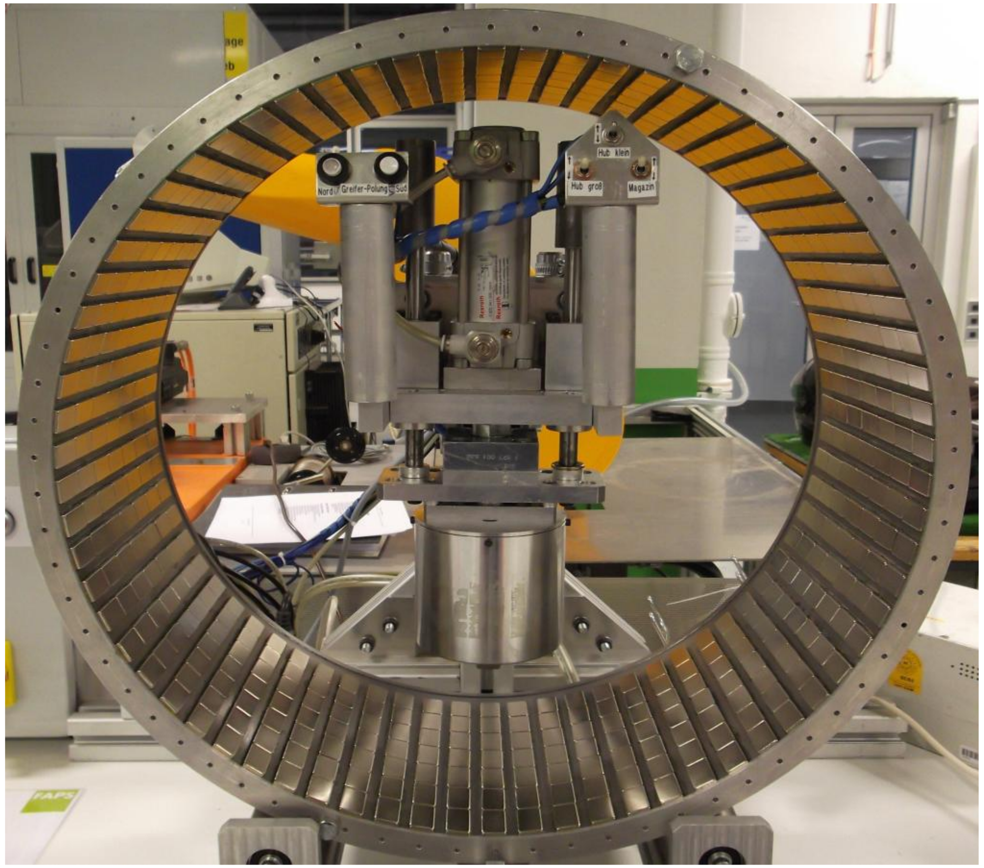
An operator-guided tool assists in the surface mounting of magnetized magnets in an external rotor generator.

General aspects of assembly:

- Precise positioning of the magnets in the cavity or on the surface results in similar running characteristics to the simulation
- High centrifugal forces require the magnets to be securely fixed in the magnet pockets or on the rotor surface
- In the case of bonding, the magnet surface often has to be cleaned and prepared by applying activators or (free) metal ions
- Complete encapsulation of rotors fixes the magnets in all degrees of freedom and protects them from environmental influences and corrosion
- The entire magnet system must be specified for the maximum operating temperature
- Observe polarity and orientation during assembly
- Sufficient rigidity of the fixtures to safely absorb the forces during installation
- Devices should generally be made of non-magnetic materials
- Temperature sensitivity of magnets is a critical factor with thermally activated or accelerated cure adhesives

Semi-automated assembly station for SPM external rotor

Magnet assembly



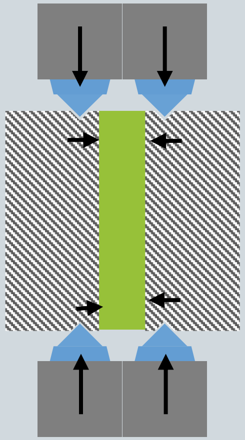
For the magnet mounting and fixing process steps, alternative joining technologies can be used.

Magnet assembly

Caulking

Force fit

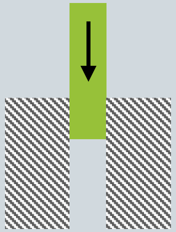
Reshaping the package with a stamp



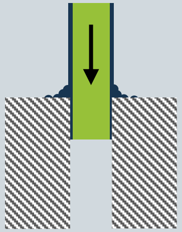
Clamping / Pressing in

Form fit / Force fit

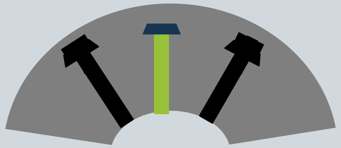
Oversize



Sacrificial layer



Additional element



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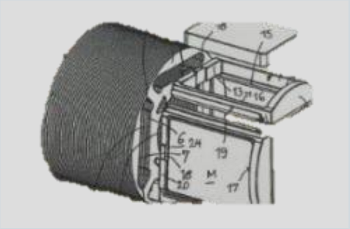
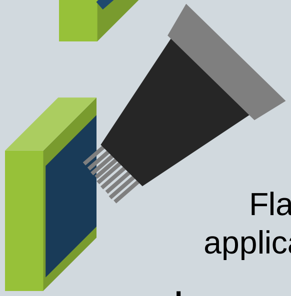


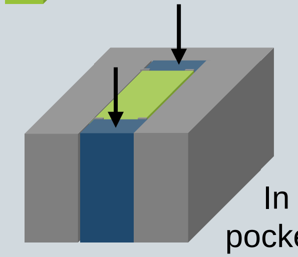
Gluing

Material locking

Point-shaped outlet



Flat application

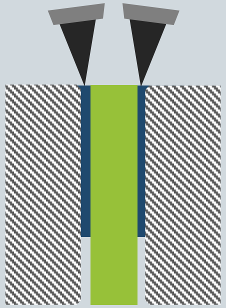


In pockets

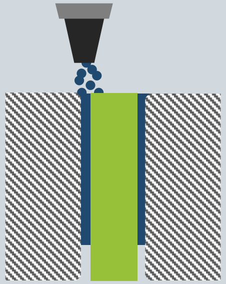
Pouring

Material locking

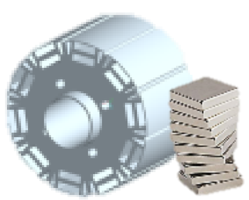
Injection molding or compression molding

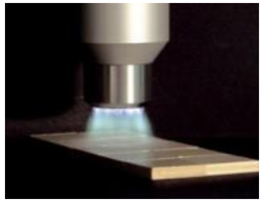


Drizzled



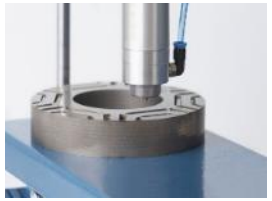
The bonding of magnets has established itself as a standard process and is therefore considered as an example.





Graphic: DELO

Graphic: DELO



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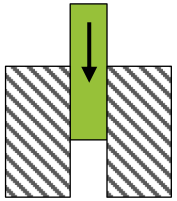
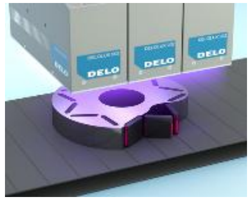


Graphic: FAPS



Graphic: DELO

Graphic: FAPS

Gluing

Sheet metal package → Preparing surfaces → Preparing the adhesive → Dosing and application → Insert magnets → Curing the adhesive → Rotor (Disc)

Magnets → Preparing surfaces

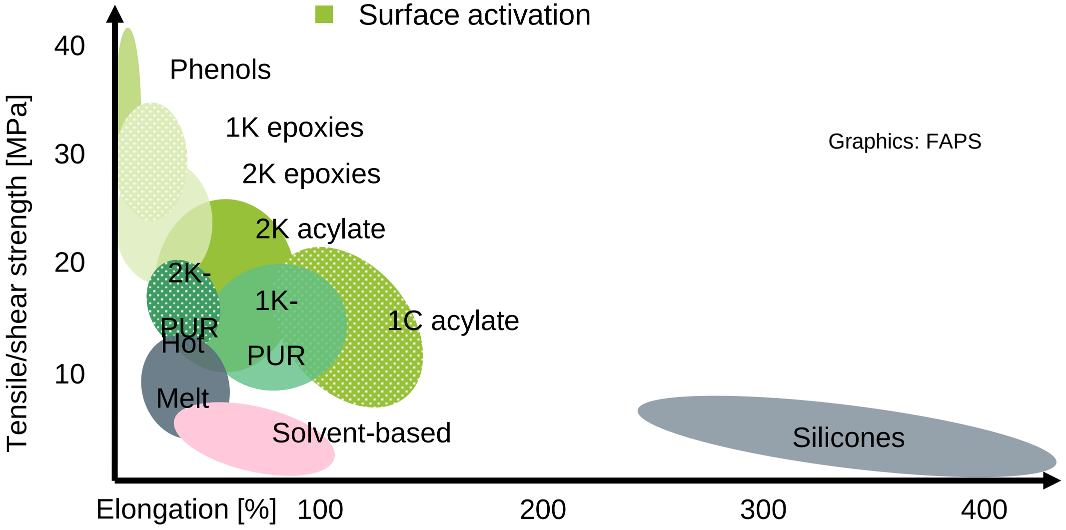
Preparing surfaces
- Degreasing the surfaces
- Mechanical roughening
- Surface activation

Preparing the adhesive
- Homogenize
- Set viscosity
- Air conditioning

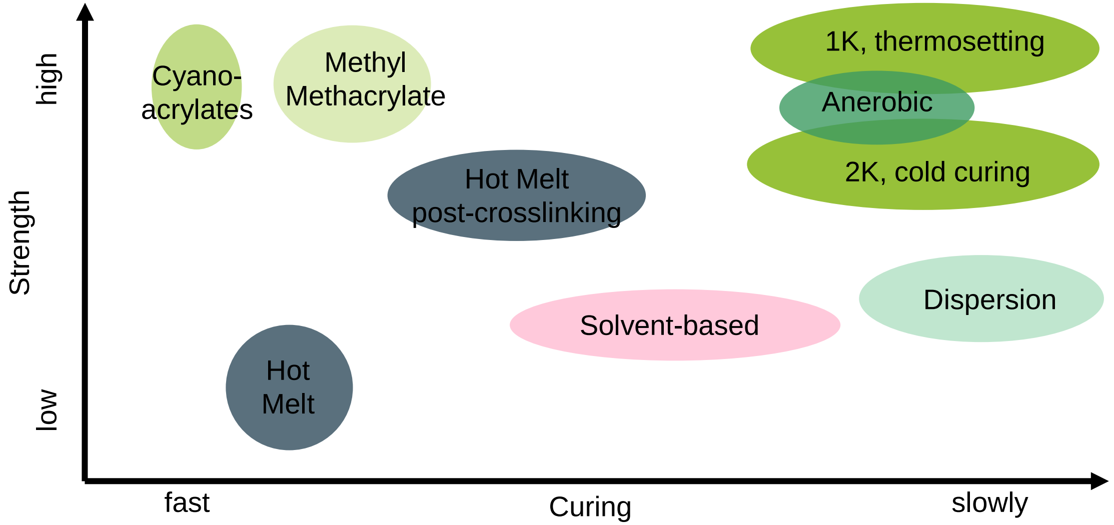
Dosing and application
- Mixing
- Application (dot, line, surface)

Insert magnets
- Positioning
- Inserting

Curing the adhesive
- Heat supply
- UV / radiation curing

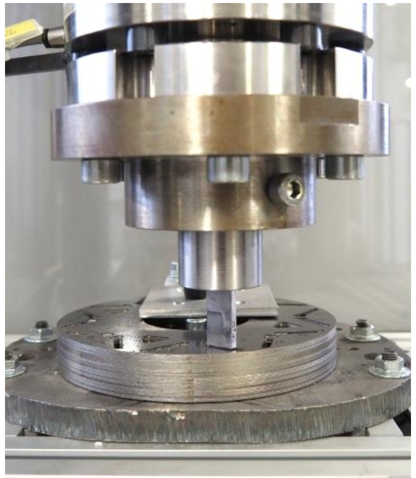


Graphics: FAPS



For the validation of adhesive bonds various methods can be used.

Adhesive application → Temperature shock test DIN 60068-2-14 → Vibration test DIN 60068-2-64 → Pressing out DIN 54452 → Gluing



Adhesive and magnets are inserted into the cavity

Parameters:

- Adhesive system
- Surface coating
- Curing conditions

Segments exposed to temperature changes

Parameters:

- Temperature change shorter than 3 min.
- -55°C - 180°C
- Holding time

Rotor is shaken in the vibration tester

Parameters:

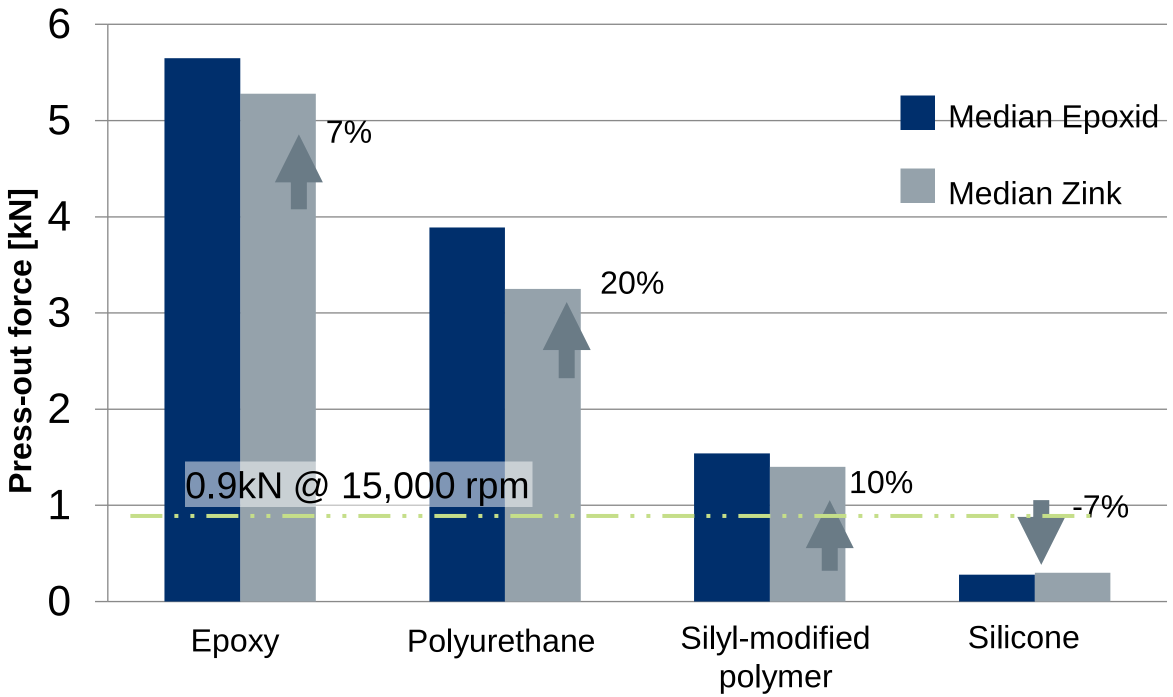
- Frequency
- Amplitude
- Duration

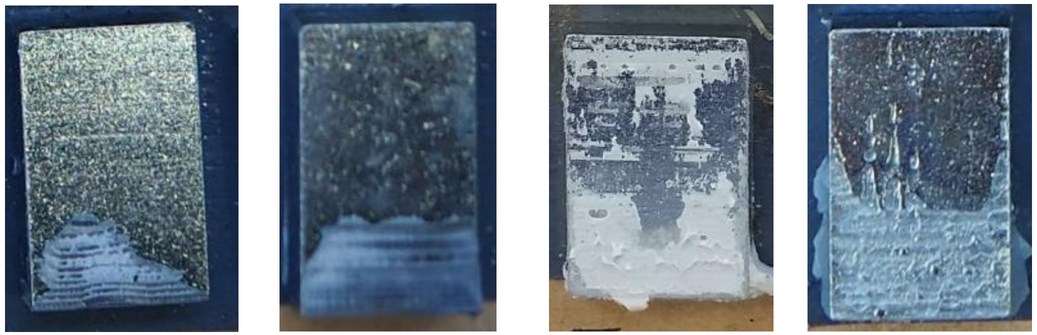
Magnets are pressed out

Output:

- Force-displacement diagram
- Elongation
- Strength

Both the adhesive systems and the surface coatings have a significant impact on the result.





Surface of the coatings

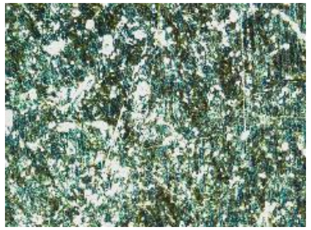
Gluing

Epoxy

R_z 9.38 µm

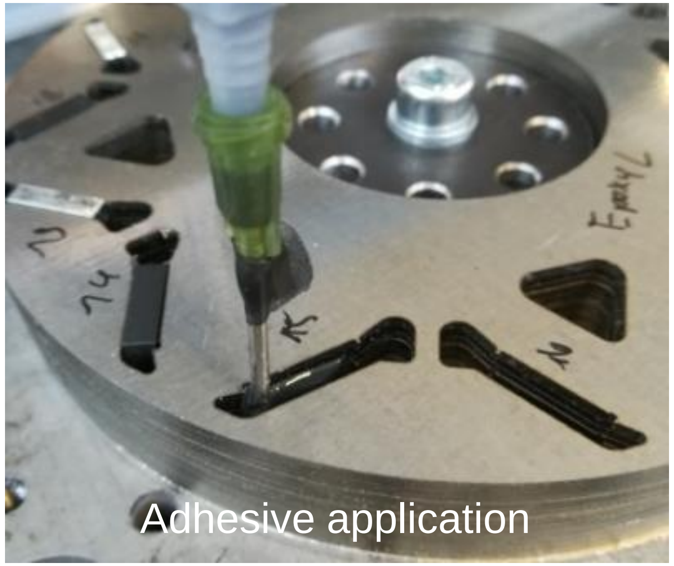
R_a 1.71 µm

Zinc



R_z 3.48 µm

R_a 0.65 µm



The challenges of bonding magnets are offset by the benefits that must be weighed against each other.

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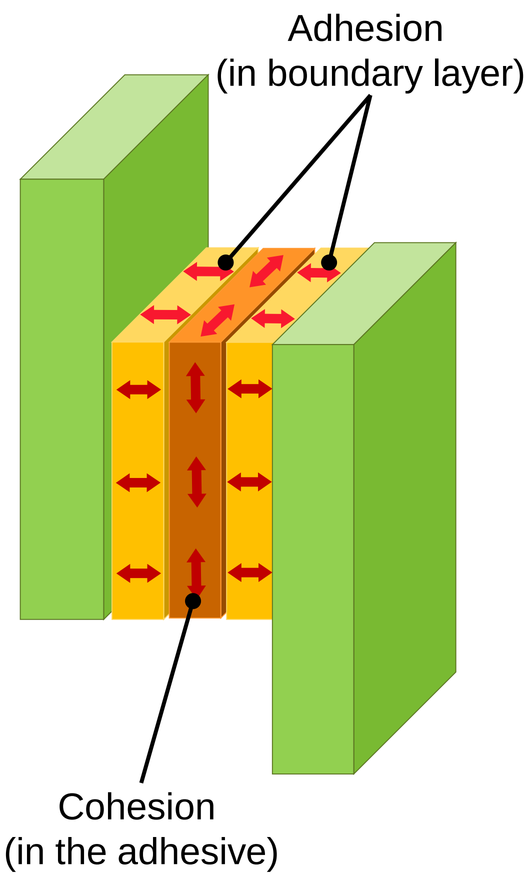
Advantages of the process

- Thermal connection to the rotor
- Full-surface connection -> homogeneous stress distribution
- Vibration damping achievable
- Oxidation and media protection
- Less mechanical stress in magnet and sheet metal
- Lower investment costs compared to injection molding
- Lower thermal load compared to injection molding
- No additional connecting parts required

Challenges of the process

- High number of process parameters
- Surface preparation required: cleanliness and activation
- Thermal stability is limited
- Complex to dismantle and recycle
- Longer process times compared to caulking
- Only destructively testable (exception: computer tomography)
- Storage and stocking of adhesives costly, system downtime problematic (curing in system)
- Emissions: Toxic vapors, skin irritation, etc.
- Comparatively unclean process (splashes)

Gluing



The integration of chemical fixing of the magnets into the assembly process leads to a considerable reduction in cycle times.

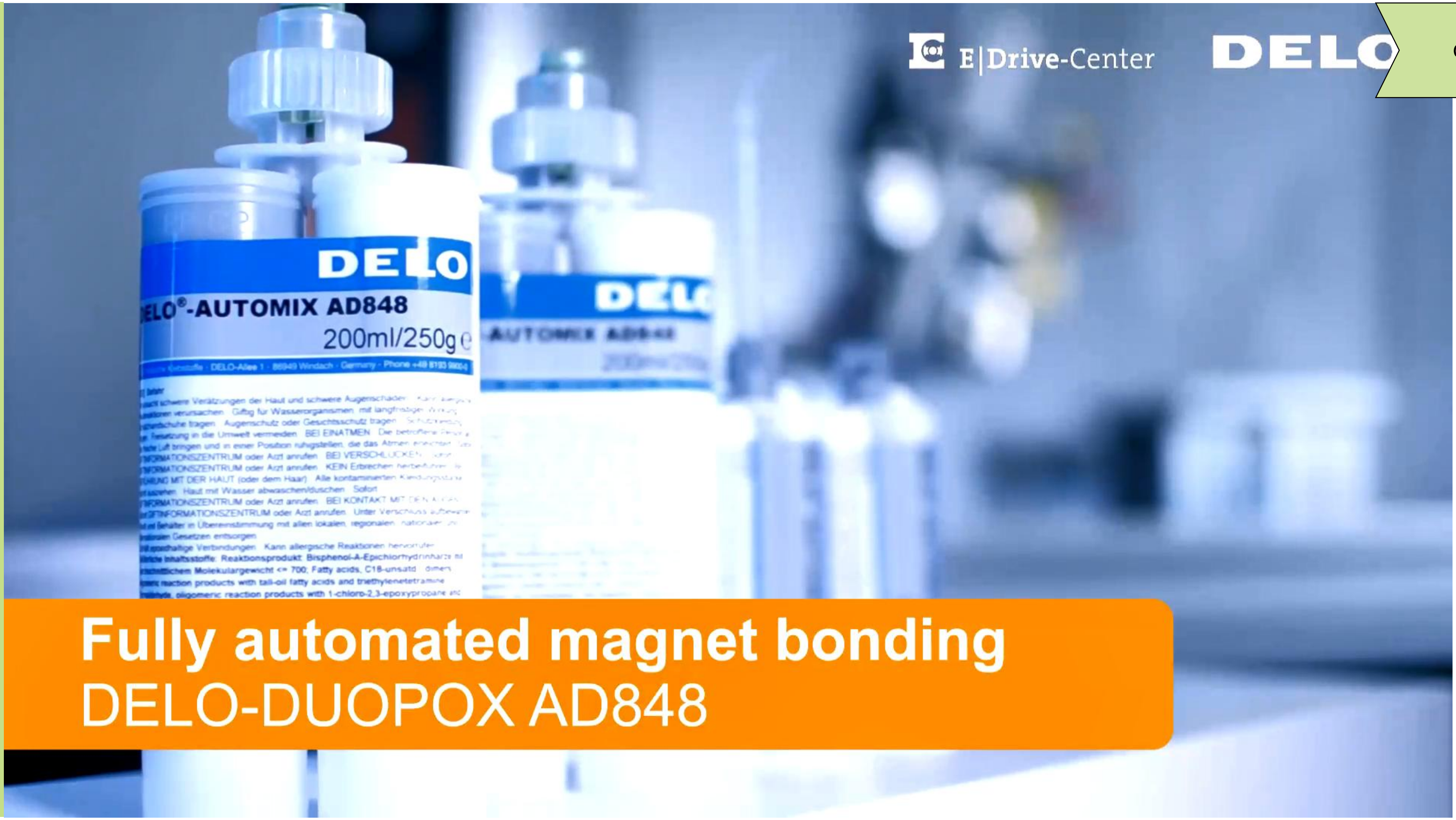
Gluing

Process

Automated adhesive application using robots

Video:
DELO/FAPS

https://youtu.be/t793NSrLNmM



The integration of chemical fixing of the magnets into the assembly process leads to a considerable reduction in cycle times.

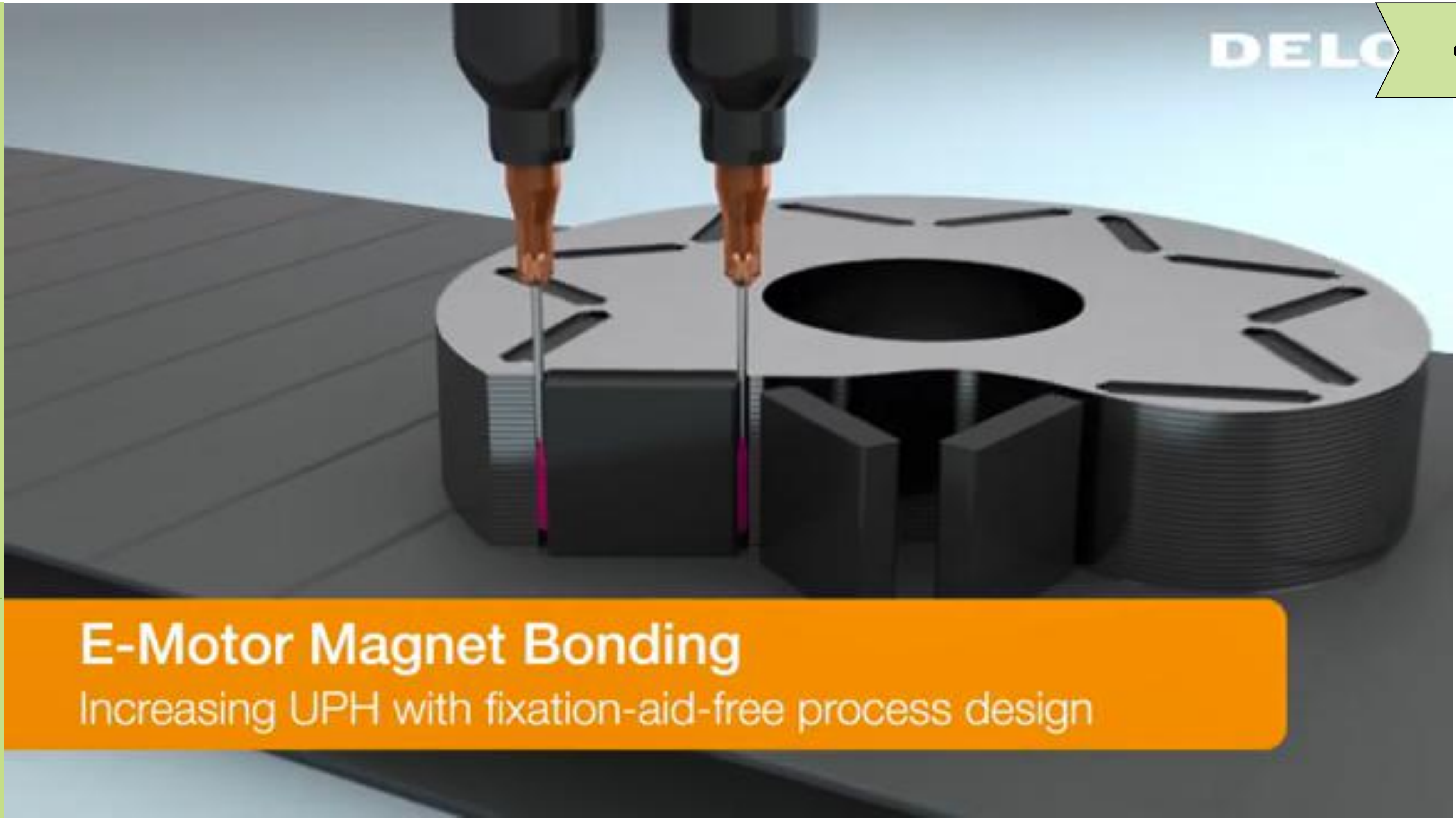
Gluing

Process

Automated adhesive application for IPM rotors

Video:
DELO/FAPS

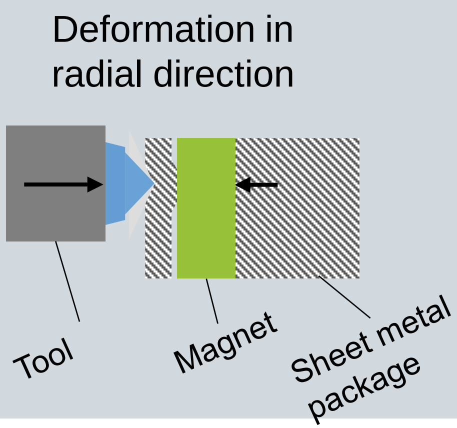
https://youtu.be/iPPj0QQiIiU

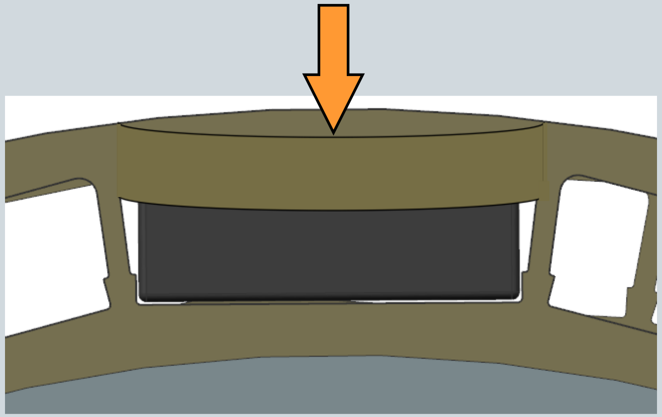


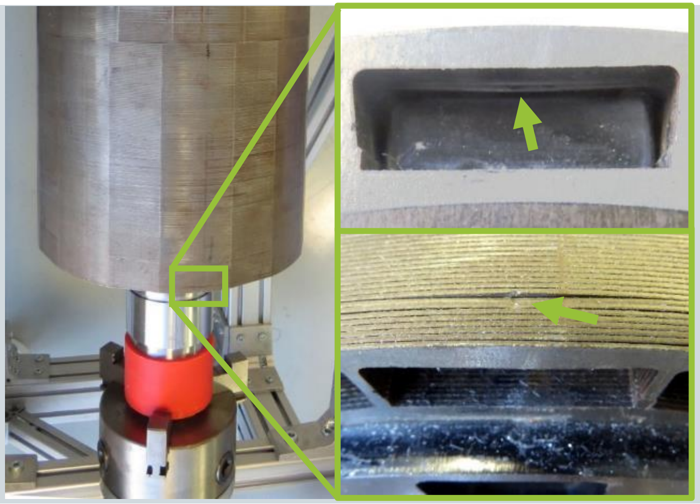
In addition to the above methods, buried magnets can also be bonded, cast, or caulked into place.

Caulking

Radial caulking direction

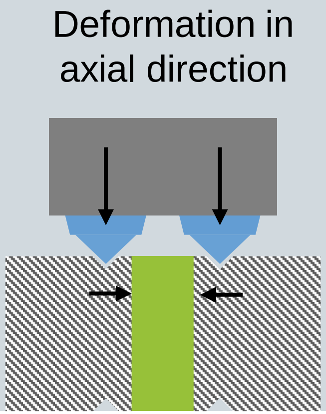


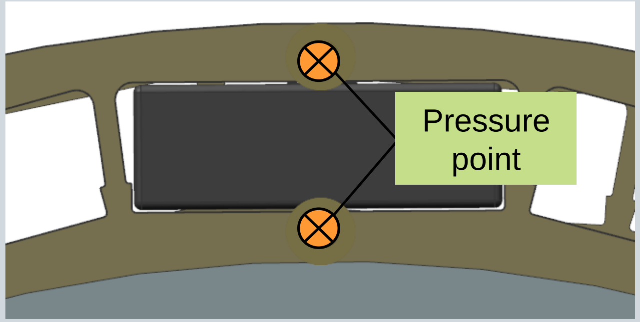


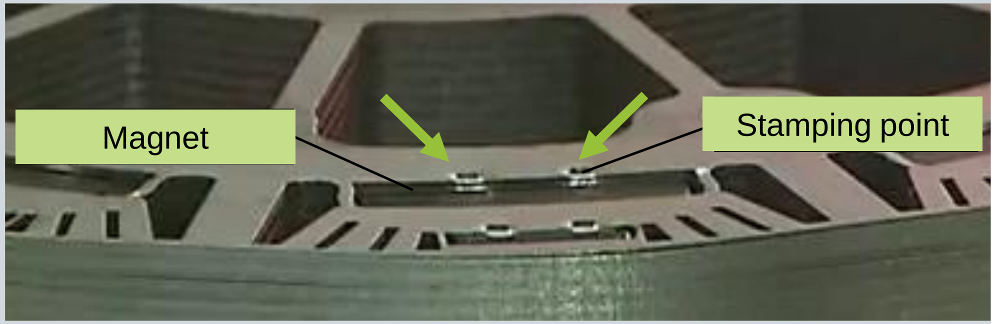


Graphics: FAPS

Axial caulking direction







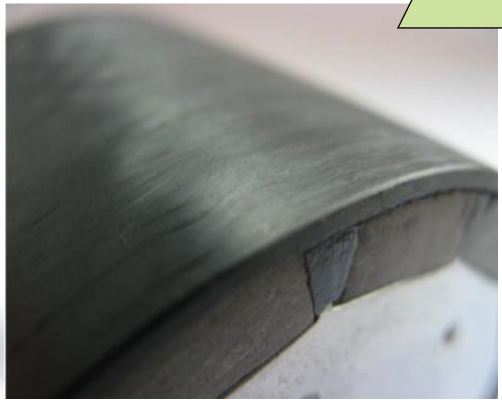
Graphic: BMW

Once bonded, surface mounted magnets are additionally secured against centrifugal and magnetic forces.

Material	Carbon steel	Aluminum alloy	GRP	CFRP
Eddy current losses	high	high	none	low
Density (g/cm)³	7.65	2.7	1.9	1.55
Tensile strength (MPa)	1400	530	1200	2200
E-modulus (GPa)	210	70	45	135
Therm. Expansion	high	high	low	low

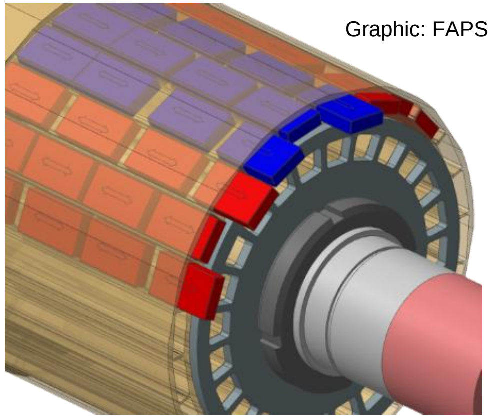
Centrifugal safety device

CFRP sleeves



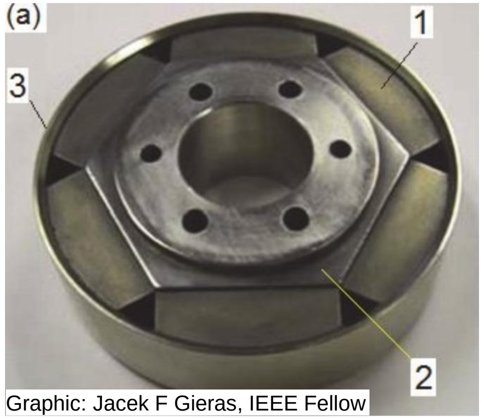
Graphic: Schunk Carbon Technology

Buried magnets secured in the sheet metal package



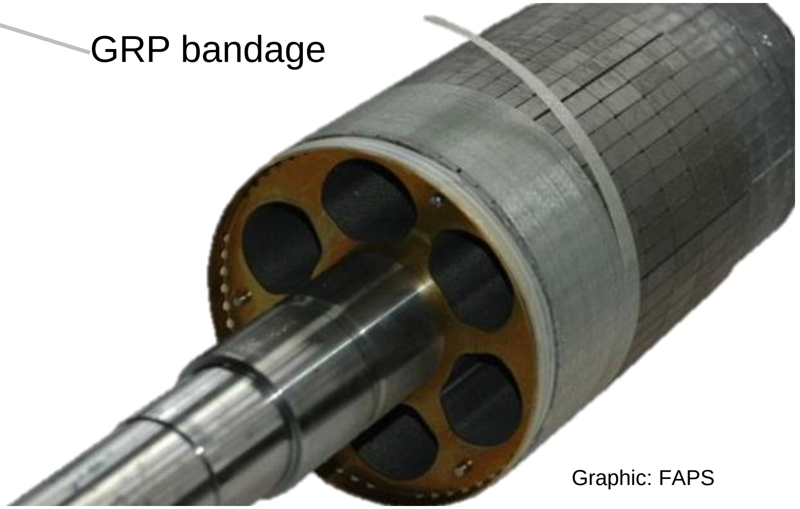
Graphic: FAPS

Steel sleeve (3)



Graphic: Jacek F Gieras, IEEE Fellow

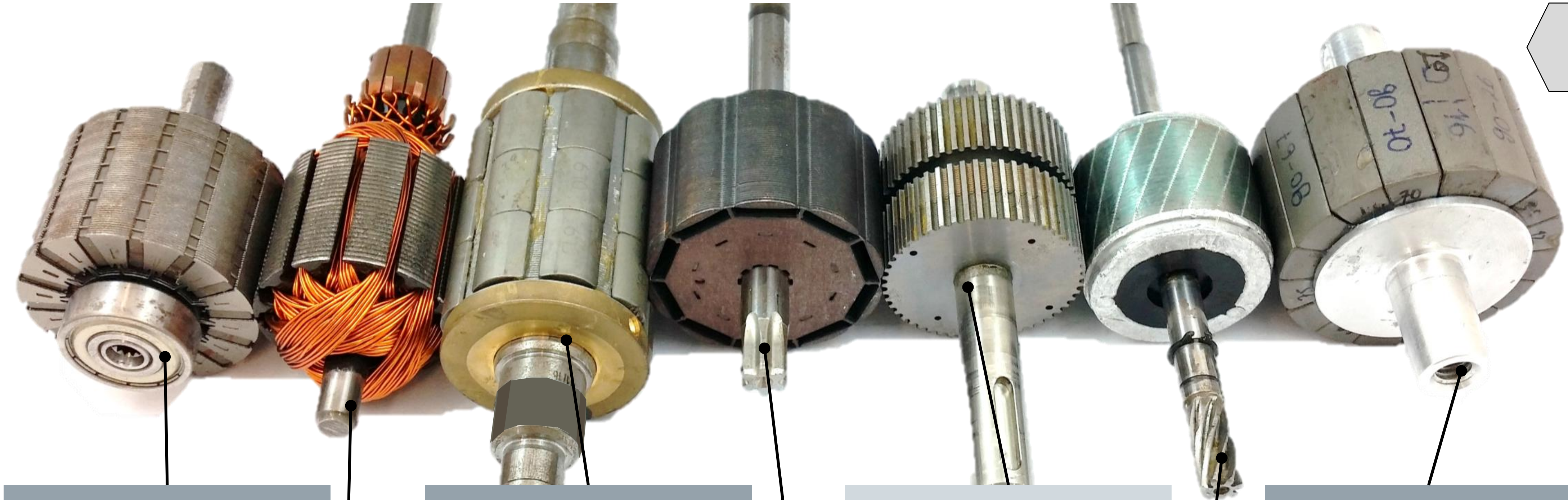
GRP bandage



Graphic: FAPS

Within the conventional design, permanent magnets are used for various motor types.

Rotor



PMSM – Flow collection system

Single-phase series-wound motor (universal motor)

PMSM - SPM; Cup-shaped PM; Balancing disks (brass)

PMSM - IPM; Trapezoidal PM

Hybrid stepper motor with PM (axially magnetized)

ASM - Short-circuit cage in die-cast with skewing angle

PMSM - (SPM) Halbach Array

Lecture unit 4 introduces the properties and manufacturing processes of magnets and explains the assembly of rotor assemblies.

Part B

Assembly of permanent magnets

- Introduction to the use of permanent magnets in electric machines
- Design of rotor assemblies and rotor topologies
- Challenges in handling magnets and magnet logistics
- Assembly and fixing magnets
- Fuses for high-speed concepts
- Qualification of connection technologies

You learned the following in part B of this lecture unit:

- Integration of rotor assembly into the motor value chain
- Rotor designs and rotor topologies
- Differences between IPM and SPM
- Generic process chain of the PSM rotor
- Handling and logistics of magnets
- Magnet assembly technologies
- Focus on adhesive bonding technology as a reference technology for magnet assembly: process steps and adhesive systems

The contents of the lecture unit can be further deepened with the following specialized literature:

Reference books

- Cassing, W. *Permanent magnets. Measuring and magnetizing technology ; with 23 tables.* Renningen: expert-Verl., 2007. contact & study. 672. ISBN 9783816925088
- Magnetic material properties. In: G. Fasching, ed. *Materials for electrical engineering. Microphysics, structure, properties.* 4th, unchanged ed. Vienna: Springer, 2005, pp. 376-453. ISBN 3-211-22133-6
- Hofmann, H. and J. Spindler. *Materials in electrical engineering. Fundamentals - Structure - Properties - Testing - Application - Technology.* 8th, updated edition. Munich: Hanser, 2018. learning books of technology. ISBN 3446458530
- Schatt, W., K.-P. Wieters and B. Kieback. *Powder Metallurgy. Technologies and materials.* s.l.: Springer-Verlag, 2007. VDI book. ISBN 9783540681120
- Coey, J.M.D., ed. *Rare-earth iron permanent magnets.* Oxford: Clarendon Press, 1996 Oxford science publications. 54. ISBN 0198517920

Other publications

- Kalyuzhnyi S.V.: "Glossary of Nanotechnology and related terms". https://eng.thesaurus.rusnano.com/
- Glöser-Chahoud, S., A. Kühn and L.A. Tercero Espinoza. *Global utilization structures of the magnetic materials neodymium and dysprosium* [online]. *A scenario-based analysis of the impact of the diffusion of electromobility on the demand for rare earths,* 2016. Available at: http://hdl.handle.net/10419/142767
- MS Schramberg: “Hystereseschleife” https://magnete.de/wissenswertes/anisotrope-magnete/hystereseschleife/

FAPS

The contents of the lecture unit can be further deepened with the following specialized literature:

Essays

- Hofmann et al, Innovative Methods for Automated Assembly and Fixation of Permanent Magnets in Electrical Machines. *ScienceDirect - 12th Global Conference on Sustainable Manufacturing,* 2014
- Gutfleisch, Controlling the properties of high energy density permanent magnetic materials by different processing routes [online]. *Journal of Physics D: Applied Physics,* 2000, **33**(17), R157-R172. ISSN 0022-3727. available at: doi:10.1088/0022-3727/33/17/201
- Lindenfels et al, Challenges in bonding processes in the production of electric motors in *Production at the leading edge of technology. Proceedings of the 9th Congress of the German Academic Association for Production Technology (WGP), September 30th - October 2nd, Hamburg 2019.* Berlin: Springer, 2019, pp. 411-420. ISBN 978-3-662-60416-8

Dissertations

- Junker, S. *Technologies and system solutions for the flexible automated assembly of permanently energized rotors with surface-mounted permanent magnets.* Dissertation, Bamberg: Meisenbach, 2007. production engineering Erlangen. ISBN 978-3-87525-259-0
- Albrecht, T. *Optimized manufacturing technologies for rotors of gearbox-integrated PM synchronous motors of hybrid vehicles.* Dissertation, Bamberg: Meisenbach, 2014. manufacturing technology Erlangen ISBN 978-3-87525-368-9
- Tremel, J. *Flexible Systems for Permanent Magnet Assembly and Magnetic Rotor Measurement.* Dissertation, Bamberg: Meisenbach, 2017 Fertigungstechnik Erlangen ISBN 978-3-87525-419-8

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THANK YOU